HQ self-service analytics — infra requirements pack

Scope: the on-prem host for the datamart (dbt-sqlserver into `POS_Analytics`) and Cube Core (Cube API, refresh worker, Cube Store), plus the DBA work on the replica `DWPOSREPL03`.
Compiled: 2026-11-05 (hosting amended the same day: one container image, no VM) from oms-react map 741 and BackOffice spec 3378.

> **For infra:** section A1 (the Nutanix request) is the ask. Everything after it is the supporting detail and the tickets it came from. Links between documents in this pack jump inside the PDF; links to files not in the pack open the file on disk and show its repo path.

Ticket status

Ticket	Repo	What it holds	State (2026-11-05)
Nutanix infra request	oms-react	The document to hand infra: one container image, internal name + Cloudflare cert, firewall lines, secrets, what to send back (rewritten 2026-11-05)	Ready to send
749	oms-react	IT + DBA approval; the full checklist	Done — DBA half verified; infra half = this request
748	oms-react	dbt-sqlserver into POS_Analytics on DWPOSREPL03	Done (decision)
750	oms-react	Replica headroom probe + DBA answers	Claimed — off-hours run and DBA answers owed
753	oms-react	Callout/donor tables on the replica	Live on subscriber — publisher counts owed
3378	BackOffice	Server-half spec	Ready
3379	BackOffice	dbt marts on the analytics host	Open — owes the single image; waits on infra running it
3380	BackOffice	Cube model over the marts	Open — blocked by 3379

Contents

A. The ask

The infra request for the analytics container (Nutanix) `oms-react/docs/research/741-nutanix-infra-request.md`

What IT and the DBA provide (full provisioning checklist) `oms-react/docs/research/741-it-dba-provisioning-checklist.md`

B. Decision tickets

749 — IT approves the hosts and service logins `oms-react/.issues/749-it-approves-the-on-prem-hosts-and-service-logins-for-the-analytics-stack.md`

748 — What builds the marts (dbt into POS_Analytics) `oms-react/.issues/748-what-builds-the-marts-views-on-the-replica-dbt-into-a-sibling-database-or-a-copy-to-a-columnar-store.md`

742 — Replica load and where derived tables live `oms-react/.issues/742-what-analytical-load-can-the-replica-carry-and-where-may-derived-tables-live.md`

C. DBA / replica tasks

750 — Replica headroom probe and the DBA checklist `oms-react/.issues/750-the-owner-runs-the-replica-headroom-probe-and-the-dba-answers-its-checklist.md`

Replica headroom and derived tables (research note) `oms-react/docs/research/741-replica-headroom-and-derived-tables.md`

741-replica-probe.sql `oms-react/docs/research/741-replica-probe.sql`

741-replica-probe.out.txt (busy-hour run) `oms-react/docs/research/741-replica-probe.out.txt`

753 — Callout and donor tables on the replica `oms-react/.issues/753-the-dba-adds-the-callout-and-donor-request-tables-to-the-replica-publication.md`

753-add-replica-articles.sql `oms-react/.issues/assets/753-add-replica-articles.sql`

D. Build tickets (BackOffice)

3378 — Server-half spec: marts, Cube model, SIS.Api doors `BackOffice/.issues/3378-hq-self-service-analytics-marts-cube-model-and-sis-api-doors-spec.md`

3379 — dbt marts into POS Analytics every 15 minutes `BackOffice/.issues/3379-the-analytics-host-builds-the-till-order-and-delivery-marts-into-pos-analytics-every-15-minutes.md`

3380 — Cube answers only declared members `BackOffice/.issues/3380-cube-answers-only-declared-members-over-the-marts-with-bilingual-labels-and-shape-metadata.md`

Analytics container — request to infra

From: the OMS BackOffice team · **Date:** 2026-11-05

Please run **one container image** that we supply, the same way you run our other apps. We build and test the image ourselves; you only run it. All its settings are built in; you set only the three secrets in §4. A ready deployment file comes with the image: `oms-analytics.k8s.yaml` (Kubernetes) or `oms-analytics.compose.yaml` (Docker).

1. Run it

Image	`oms-analytics:<version>`, linux/amd64. Delivered the same way as our other apps' images.
Size	**4 vCPU, 16 GB RAM** (may later grow to 8 vCPU / 32 GB).
Instances	**Exactly one.** Never scale it out or run two at once.
Disk	**None.** Nothing needs to persist; it rebuilds its cache on start.
Restart	Always restart on failure. A restart is always safe.
Port	The container listens on **4000** only.

2. Name and certificate

- An **internal-only** DNS name (not published on the internet), e.g. `oms-analytics.<internal domain>`.
- A Cloudflare certificate on **443**, routed to container port **4000**.

3. Network

From → to	Port	Rule
Production SIS.Api and staging SIS.Api → the container	443	allow
Your monitoring → the container	443	allow (health check only)
The container → `DWPOSREPL03` / 172.23.27.47	1433	allow
Anything else in, and the container → the internet	—	**deny**

4. Secrets (environment variables, set at deploy time)

`CUBEJS_DB_PASS` · `DBT_ENV_SECRET_PASSWORD` · `CUBEJS_API_SECRET`

We give you the values in person or through your secret store — never by email or ticket. The container refuses to start (exit code 64) if one is missing.

5. Health checks

Path	Use it for	Healthy
`GET /livez`	liveness (restart if failing)	`200`
`GET /health`	**monitoring and alerts**	`200`; `503` means the service or its data is stale. The JSON body says which.

Please alert **the OMS BackOffice team** when `/health` returns 503 for more than 15 minutes.

6. What we need from you

- **Back on this request:** the DNS name, the source IPs of production and staging SIS.Api, and how we hand you a new image version.
- **Running:** read access to the container's logs, and a way to see it running and restart it (or a named contact who can). **No shell access is needed.**

741 / 749 — What IT and the DBA provide before the analytics build starts

The checklist for IT approves the on-prem hosts and service logins for the analytics stack. It covers the stack the map settled: Cube Core with Cube Store (743, 756), and dbt-sqlserver into `POS_Analytics` (748, 757). SIS.Api is the only caller (746, 747). Metabase is **not** in v1, so nothing here provisions it.

Written 2026-11-04. Every sizing figure and every "recommended" choice is the agent's call. The owner may overrule any of them. Facts measured on the replica that day are marked **(measured)**.

> **Amended 2026-11-05 (owner ruling): §1 and §2 are superseded.** Infra already runs our apps as containers on Nutanix, so there is no new VM. The stack ships as **one image** (Cube API with its refresh worker, Cube Store on localhost only, and dbt with its schedule inside, overlap guarded by a lock file). Infra runs it as one instance, with an **internal-only** DNS name and a Cloudflare certificate on 443, SIS.Api in, `DWPOSREPL03:1433` out, no disk, no internet. **D8 is (a):** pin the replica's current certificate; the internal-CA certificate is no longer asked. The live ask is 741-nutanix-infra-request.md. §3 (the DBA half) stands as done.

0. What the build needs, on one line

One Linux amd64 VM with Docker runs four containers. It reaches `DWPOSREPL03:1433` and is reached only by SIS.Api. It needs a sibling DB, `POS_Analytics`, with two SQL logins. It adds **no** internet egress.

```
browser ──► SIS.Api (IIS, existing) ──HTTPS 443──► analytics host ──TDS 1433──► DWPOSREPL03
                │                                    cube-api · cube-refresh · cubestore · dbt (cron)
                └──HTTPS──► model API (734 D9, unchanged)
```

1. IT — the analytics host

#	Item	Recommended	Why
H1	**One VM, Linux, x86-64 (amd64)**	Ubuntu 24.04 LTS, or IT's standard RHEL 9	Cube Store has **no arm64 image** (756). The spikes never tried Windows-native Cube Store, so a Windows host is not an option.
H2	**Sizing**	**8 vCPU, 32 GB RAM, 150 GB SSD** (40 GB OS, 110 GB `/var/lib/docker` and Cube Store data). The floor is 4 vCPU and 16 GB.	Cube Store keeps rollups in memory and on local disk. dbt jobs are short (8–11 s cold, 757). The rollups are small: about 1,561 stores **(measured)** × days × channel. The headroom is for the 30 s auto-refresh boards (755) hitting Cube's cache.
H3	**Container runtime**	Docker Engine with the compose plugin, from the distro or Docker's repo. Podman with compose is acceptable if it is IT's standard.	The four services are a single compose file. The dbt overlap guard needs `docker run --name` (757).
H4	**Scheduler**	Host `cron` or a systemd timer that starts the dbt container: intraday every 15 minutes, nightly at 03:00 Riyadh time (748).	It keeps Python and our jobs off `DWPOSREPL03` and out of the DBA's SQL Agent (748).

#	Item	Recommended	Why
H5	**Images reach the host offline**	We build and pin the images by digest, `docker save` them, and IT copies the tarballs in.	The host then needs **no** registry, PyPI or `packages.microsoft.com` egress. If IT prefers it, allow pull-only egress to `registry-1.docker.io`, `mcr.microsoft.com` and `ghcr.io` during install, and close it afterwards.
H6	**Telemetry off**	`CUBEJS_TELEMETRY=false` and `CUBESTORE_TELEMETRY=false`. dbt `send_anonymous_usage_stats: false`.	With telemetry off and offline images, the host makes **zero** outbound internet calls.
H7	**Secrets on the host**	One env file per service, mode `0600`, owned by the service account. The files hold `CUBEJS_DB_PASS`, `CUBEJS_API_SECRET` and `DBT_ENV_SECRET_PASSWORD` (757).	No secret goes in the repo or the image.
H8	**Backups**	No VM backup is required. An image-level snapshot after install is enough.	Cube Store is rebuilt from `POS_Analytics`. The model comes from the BackOffice repo. Saved views and boards live in **SIS.Api's** store (755), not on this host.
H9	**Two environments, one host**	Staging SIS.Api and production SIS.Api both call the **same** analytics host, which reads the production replica.	There is only one replica, and it is production's. v1 is aggregate-only, HQ-only, with no PII member (745), so staging seeing production aggregates is acceptable. The owner may overrule this and ask for a second host.

2. IT — network paths (firewall lines)

#	From → to	Port	Notes
N1	**SIS.Api host(s)** (production behind `oms-backoffice.al-dawaa.com`, plus staging) → **analytics host**	**TCP 443**	Cube's API, behind a TLS-terminating reverse proxy container (nginx or Caddy) with an **internal-CA certificate**. Only SIS.Api's source IPs are allowed. Each call also carries a short-lived HS256 JWT signed with `CUBEJS_API_SECRET`. SIS.Api holds that secret in its server secret config, as 734 D9 does for the model key.
N2	**Analytics host** → `DWPOSREPL03` **(172.23.27.47)**	**TCP 1433**	Both dbt (reads `POS`, writes `POS_Analytics`) and Cube (reads `POS_Analytics`) use it. The connection is encrypted (see D8).
N3	**No browser → analytics host**	—	Users reach only SIS.Api (747). Port 4000 (Cube) and Cube Store's ports are never published off the host. Cube Store has **no auth** and stays on the compose network (756).
N4	Admin SSH to the analytics host	TCP 22	From IT's usual jump host only.
N5	**Model egress**	—	**Unchanged.** It stays the one line 734 D9 approved: the SIS.Api host → the model API. The analytics stack adds **no** egress line. ⚠ This is a 734 matter, not this map's: D9 names only the Anthropic line, but 734 D4 also ships a Gemini connector. Switching it on would need a second line from SIS.Api → `generativelanguage.googleapis.com`.
N6	NTP and DNS	as standard	The freshness gate compares UTC times (757).

3. DBA — `DWPOSREPL03`

Measured 2026-11-04: **Enterprise Edition, SQL Server 2019** (15.0.4470.1). The sibling DBs `ZC` and `BusinessMonitor` already exist, which is a precedent for item D1 **(measured)**.

#	Item	Recommended
D1	**Create `POS_Analytics`** on `DWPOSREPL03`, outside the publication.	Compatibility level **150**. The DB survives a reinitialisation of `POS` (748).
D2	**Size budget**	Start the data file at 10 GB and the log at 2 GB, autogrow by 1 GB, **capped at 40 GB**. Estimate: `fct_delivery` has one row per delivery. About 7.1M `DeliveryHeader` rows exist all-time **(measured)**, so it is around 1–3 GB with its indexes. The daily marts are under 1 GB. The cap leaves several years of growth.
D3	**Recovery and backups**	`SIMPLE` recovery. No backups are required, because the marts rebuild from `POS`, which holds the full history. The DBA may still take a weekly full.
D4	`READ_COMMITTED_SNAPSHOT ON` for `POS_Analytics`	**Required.** A NOLOCK reader saw a ~5 ms gap mid-swap (757). It sits beside `POS`'s own RCSI, which is already on (750).
D5	**Login `svc_analytics_dbt`**, SQL authentication, policy on	`db_owner` on `POS_Analytics`. **`db_datareader` on `POS`**, with nothing written there (757: 107 reads, 0 writes). Why a role, not table grants: a reinit whose `@pre_creation_cmd` is `drop` re-creates the articles, and per-table `GRANT SELECT` would vanish with them. Role membership survives.
D6	**Login `svc_analytics_cube`**, SQL authentication, policy on	`db_datareader` on `POS_Analytics` **only**. It gets **no user in `POS`**, so the interactive engine cannot touch the replica (748).
D7	**Resource Governor** (asked, not required)	One workload group per login. `svc_analytics_cube`: **`MAX_DOP 4`**, because unbounded parallelism cost 3× the CPU (757). `svc_analytics_dbt`: `MAX_DOP` 2 and `REQUEST_MAX_MEMORY_GRANT_PERCENT` 25. This needs a classifier function, so it is the DBA's call (750 item 1).
D8	**TLS trust**	The sanctioned `sqlcmd` shape uses `-C`, which suggests the server certificate is self-signed. Choose one: (a) export the server certificate's public part for the containers' trust store (no restart; it breaks when the certificate is renewed), or (b) install a certificate issued by the internal CA, which needs a SQL service restart. Either way the build **never** ships `trustServerCertificate` (756, 757). Cube needs a `driverFactory` for this, already written.
D9	**Auto-created statistics** (a note, nothing to do)	The dbt reads make the engine auto-create `_WA_Sys_*` statistics on `POS` tables (757 saw 24). They are harmless, and a reinit drops them.
D10	`VIEW DATABASE STATE` on `POS_Analytics` for `svc_analytics_cube` (optional)	For tuning only (750 item 10). It does not gate v1.
D11	Area 4's source	The callout and donor articles are their own ticket, 753. They gate v1 area 4 only, not this checklist.

4. Dev-side items this surfaced (for the ADR, not IT)

- ⚠ **The replica is SQL Server 2019, and both spikes ran on 2022 CU27.** No 2022-only function (`GREATEST`, `LEAST`, `DATETRUNC`, `DATE_BUCKET`, `GENERATE_SERIES`, `IS DISTINCT FROM`) appears in either spike's SQL. `PERCENTILE_CONT`, `delete+insert` and RCSI are all 2019 features, so nothing is known to break. The ADR should still pin **compat level 150** and run the build's dbt and Cube smoke tests on a SQL 2019 container. Cube generates its SQL itself, so its date truncation is the part to watch.

- **The dbt image** was built only for arm64 (757). The build must produce and smoke-test the amd64 image. It has no arch-specific step.
- **SSO for a BI tool:** not v1. It applies only to a later Metabase Pro tier (746, 747).

5. What the owner records back on the ticket

- the host name or IP, its OS, and the size IT gave it;
- the URL SIS.Api calls (`https://<analytics-host>/cubejs-api/v1`) and which SIS.Api hosts are allowed;
- whether the images reach the host offline (H5) or by temporary egress;
- `POS_Analytics` created: compat level, the RCSI state, the size cap;
- the login names, **never their secrets**, plus where the secrets are kept;
- the Resource Governor groups (yes or no), and the TLS choice (D8 a or b);
- any condition IT or the DBA attached.

type	wayfinder-ticket
wayfinder	task
map	741
status	done
blocked-by	746, 747, 748

749 — IT approves the on-prem hosts and service logins for the analytics stack

Question

Once the pieces are chosen, what must IT and the DBA provide before the build can start?

Hand the owner a precise checklist covering: - the host or hosts: VM sizing and the container runtime; - the service logins on the replica, or a sibling database, and their least-privilege grants; - the network paths: from the BI tool and the semantic layer to the replica, and from SIS.Api to the semantic layer; - the SSO registration for the BI tool; - whether the model egress stays the one line 734 D9 approved.

Sharpened by What builds the marts (2026-11-04). The checklist must also carry these items: - **Database:** `POS_Analytics` on `DWPOSREPL03`, the sibling DB (DBA item 9), with its size budget. - **Two logins:** - `svc_analytics_dbt`: `db_owner` on `POS_Analytics` and `db_datareader` on `POS`; - `svc_analytics_cube`: `db_datareader` on `POS_Analytics` only. - **The dbt scheduler:** a container on the same analytics host as Cube and Cube Store. - **Network path:** analytics host → `DWPOSREPL03:1433`.

The "SSO for the BI tool" bullet above applies only to the later Metabase Pro tier (746/747). It does not gate v1.

Resolved when IT has approved, or has named its conditions. The answer records the hosts, the login names (never their secrets) and the URLs that the spec relies on.

Progress

- **2026-11-04, the checklist is ready; the ticket is not resolved.** It is 741-it-dba-provisioning-checklist.md. It has 9 host items, 6 network lines and 11 DBA items, plus what to record back. The sizing and the "recommended" choices are the agent's calls, and the owner may overrule them:
- **Host:** one Linux amd64 VM with Docker, 8 vCPU / 32 GB / 150 GB.
- **Images:** they reach the host offline, so the host has **zero** internet egress.
- **One host for both environments:** staging and production SIS.Api both call it.
- **TLS:** SIS.Api → the host on 443, with an internal-CA certificate and a JWT.
- **`POS_Analytics`:** compat level 150, RCSI on, `SIMPLE` recovery, capped at 40 GB.
- **Why the dbt login keeps `db_datareader` on `POS`:** a reinit drops per-table grants.
- **The certificate:** choose between pinning the replica's self-signed certificate and installing an internal-CA one.
- **Measured on the replica through the sanctioned door:**
- it is **SQL Server 2019** (15.0.4470.1, Enterprise), not 2022 as both spikes assumed;
- `ZC` and `BusinessMonitor` are sibling DBs, a precedent for `POS_Analytics`;
- about 7.1M `DeliveryHeader` rows exist all-time;
- there are 1,561 stores.

- The spikes' SQL uses no 2022-only function. The ADR pins compat level 150 and a smoke test on SQL 2019 (§4 of the checklist).
- **Flagged for spike 734, not this map:** D9 approves only the Anthropic egress line, but D4 ships a Gemini connector.
- **Next (HITL):** the owner puts §1–§3 to IT and the DBA, then records §5's answers here. Any session can then close the ticket.
- **2026-11-04 22:3x, the DBA half has started; the ticket is not resolved.** The owner is acting DBA until a dedicated one is hired, so §3 is the owner's to do. Checked through the sanctioned door:
- `POS_Analytics` **exists** on `DWPOSREPL03`, created 2026-11-04 22:27:27. It is compat level **150** and **RCSI on**, as §3 asked.
- **Its recovery model is `FULL`; §3 asked for `SIMPLE`.** The marts can be rebuilt from `POS`, so nothing needs point-in-time restore. Under `FULL` with no log backups, the dbt `delete+insert` churn (every 15 minutes) grows the log without bound. Either switch it to `SIMPLE` or schedule log backups. **Settled the same evening: the owner switched it to `SIMPLE`, re-verified.**
- **Not verifiable from the `read` login:** it has no `VIEW ANY DEFINITION` and no access to `POS_Analytics`. So the two `svc_analytics_*` logins, their grants, and the 40 GB cap stay owed from the owner.
- **Still owed from IT:** the analytics host (§1), the firewall lines (§2), and §5's answers.
- 753's tables are still not on the subscriber. The only new tables since 2026-11-03 are a `Julib*` snapshot at 09:11 on 2026-11-04.
- **2026-11-04, later, the logins verified by logging in as each; the ticket is not resolved.**
- `svc_analytics_dbt:` `db_owner` on `POS_Analytics`. On `POS` it is `db_datareader` with no writer or owner role, and sees all 274 tables. This matches §3.
- `svc_analytics_cube:` `db_datareader` on `POS_Analytics`, with no write and no `CREATE TABLE`. It has **no access to `POS`**. This matches §3.
- **Where the secrets are kept:** in the owner's gitignored `.passwords/POS_Analytics.md` (`.gitignore:163`, never tracked), until the analytics host's env files exist (H7).
- **Gap 1, no size cap.** The data file is 1 GB with 1 GB growth and `max_size` unlimited; §3 asked for a 40 GB cap. The log has the default 2 TB cap.
- **Gap 2, `MAXDOP` 0.** 757 caps Cube's login at `MAXDOP` 4 or less. A database-scoped `MAXDOP 4` on `POS_Analytics` covers it, since only the marts are read there.
- The database is still empty (0 tables), as expected before the first `dbt build`.
- **2026-11-04, later, both gaps closed by the owner and re-verified.** The data file is capped at 40 GB and the log at 8 GB, and the database-scoped `MAXDOP` is 4. **§3 (the DBA half) is now complete:** `POS_Analytics` is compat 150, RCSI on, `SIMPLE`, capped, and both logins hold exactly their §3 grants. **Only IT's half is still owed:** the analytics host (§1), the firewall lines (§2), and its §5 answers: host or IP, OS, size, the URL and the allowed SIS.Api hosts, offline images or not, the TLS choice, and any conditions.

Answer

Approved, 2026-11-04. The owner speaks for IT and is acting DBA, and ruled the ticket solved. Infra provisions from 741-nutanix-infra-request.md.

- **The host:** one **Docker host VM on Nutanix AHV**: Linux amd64, 8 vCPU, 32 GB RAM, 150 GB SSD, with a static IP, an internal DNS name and an internal-CA certificate. It is **not** the Kubernetes platform *(agent; the request's last section maps it onto Kubernetes if infra requires that)*. Images come from an internal registry, or else as offline tarballs. The VM has no internet egress.
- **The network:** production and staging SIS.Api → the VM on 443, the VM → `DWPOSREPL03:1433`, and SSH from the jump host. Nothing else gets in. **Model egress is unchanged:** it is the one 734 D9 line from SIS.Api.
- **The database, done and verified on the replica:** `POS_Analytics` is compat 150, RCSI on, `SIMPLE`, capped at 40 GB data and 8 GB log, with `MAXDOP` 4 database-scoped.

- **The logins, verified by logging in as each:** `svc_analytics_dbt` is `db_owner` on `POS_Analytics` and `db_datareader` on `POS`; `svc_analytics_cube` is `db_datareader` on `POS_Analytics` only. Their secrets are in the owner's gitignored `.passwords/`, and move to the host's env files (H7).
- **Resource Governor: not in v1** *(agent)*. The scoped `MAXDOP` covers 757's CPU risk.
- **SQL TLS (D8): (b)** *(agent)*, an internal-CA certificate on `DWPOSREPL03` in a maintenance window. Until then the build pins the current certificate. It never ships `trustServerCertificate`.
- **SSO:** none in v1. It applies only to the later Metabase Pro tier.
- **The URL the spec relies on:** `https://<analytics-host>/cubejs-api/v1`. Infra's name for the host, its IP and the SIS.Api source IPs are **build inputs**, not decisions. They get filled in when infra hands the VM over, and the build tickets record them. Nothing in the spec or the ADR waits on them.

Amended 2026-11-05 (owner, grilled)

- **No VM.** Infra already runs our apps as containers on Nutanix. We build the image; infra runs it. The "host" and "Why not Kubernetes" bullets above are superseded.
- **One image**, not four containers: Cube API with its refresh worker, Cube Store on localhost only (its missing auth never reaches a network), and dbt with its 15-minute and 03:00 schedule inside, overlap guarded by a lock file instead of a fixed container name. **Exactly one instance**, no persistent disk (Cube Store is a cache, rebuilt on restart), 4 vCPU / 16 GB to start.
- **The name is internal-only**, with a Cloudflare certificate terminated by infra's platform on 443 → container port 4000. Only production and staging SIS.Api reach it. SIS.Api trusts the Cloudflare chain as-is, so no CA chain is installed.
- **Network:** SIS.Api → 443, the container → `DWPOSREPL03:1433`, nothing else. No SSH line.
- **D8 is (a):** pin the replica's current certificate, re-copied at its renewal; never `trustServerCertificate`. The internal-CA certificate (b) is no longer asked.
- **Secrets:** three env vars set by infra at deploy time (`CUBEJS_DB_PASS`, `DBT_ENV_SECRET_PASSWORD`, `CUBEJS_API_SECRET`), never in the image.
- The live ask is the rewritten 741-nutanix-infra-request.md. The URL the spec relies on is now `https://<internal-name>/cubejs-api/v1`.

Source: oms-react/.issues/748-what-builds-the-marts-views-on-the-replica-dbt-into-a-sibling-database-or-a-copy-to-a-columnar-store.md

type	wayfinder-ticket
wayfinder	grilling
map	741
status	done
blocked-by	742, 743, 745, 750 (750 waived by the owner, 2026-11-04 — settled provisional on it)

748 — What builds the marts: views on the replica, dbt into a sibling database, or a copy to a columnar store?

Question

Given the replica's headroom (742), the semantic layer's caching (743) and the v1 subject areas (745), what is the **pipeline**?

The options: 1. **No pipeline:** the semantic layer queries the replica directly, with only its own pre-aggregations. 2. **dbt into a sibling database:** dbt (`dbt-sqlserver`) or SQL Agent jobs build marts into a sibling database on the replica's instance. 3. **A columnar copy:** an off-replica copy, for example ClickHouse or DuckDB, fed by a scheduled extract or CDC.

Settle: - the refresh cadence per subject area. 734 D2's nightly rollup, plus a same-day partial every 15 minutes, is the precedent; - who owns and runs the jobs; - what happens when the replica is reinitialised; - whether 734's `sales_summary` rollup becomes one of these marts rather than a second copy.

Answer

Settled with the owner, 2026-11-04. 750 was still half done: the busy-hour run was in, but the off-hours run and the DBA's answers were owed. The owner chose to settle now, with 750's open items as **re-open triggers** (below) rather than blockers.

The owner's four rulings

1. **Settle now**, provisional on 750.
2. **The pipeline is option 2: dbt into a sibling database.** - `dbt-sqlserver` (dbt Core, Apache-2.0) builds marts into `POS_Analytics`, a new database beside `POS` on `DWPOSREPL03`. `ZC` and `BusinessMonitor` are the precedent (742 §3a). - Cube (743/746) reads the marts. It no longer reads the raw till and order tables for aggregates. - This overrules my recommendation of Cube Store rollups only, with no pipeline. - Options 1 and 3 are out for v1. Option 1 comes back only as the fallback named below.
3. **Delivery-speed percentiles are computed live and capped at 92 days.** - Each query computes its median or p90 exactly: `PERCENTILE_CONT`, SQL Server's twin of `PERCENTILE.INC`, the method `DispatchScorecardMath.Percentile` uses. - Results are cached for 15 minutes. - The semantic model refuses a percentile measure over a window longer than 92 days. Counts and shares have no such cap.
4. **Freshness:** today's slice refreshes every 15 minutes, and the last three days are rebuilt nightly. This is 734 D2's precedent, applied to every area.

What follows from them (my rulings, overrulable)

The marts. Each mart is incremental, keyed by business date, and replaced a whole day at a time.

Area	Mart (grain)	Feeds
1 Till sales	`mart_till_sales_daily` (date × store × country × NPHIES-linked flag): additive counts and SAR sums for every till measure in the 745 catalogue. Shares are stored as numerator and denominator, never as a ratio	`till_sales` … `nphies_linked_sales`; 734's `sales_summary`
2 Orders	`mart_orders_daily` (date × store × channel group × document type)	`orders`
3 Delivery speed	`fct_delivery`: one row per delivery, holding its first-ever `DCRT` / `DRDY` / `DOFD` / `DDLR` times (ADR 0081), the order's `EntryTime`, its store and its channel. About 6M rows a year. Plus `mart_delivery_daily`, which holds the counts and the same-day numerator	counts from the daily mart; **percentiles computed live over `fct_delivery`**
4 Callouts, donors (after 753)	`fct_callout` and `fct_donor_request` (one row per item), plus their daily count marts	the same pattern: counts from the daily marts, medians computed live over the fact tables. Parity with the scorecard is the acceptance check

- **What "live" means under ruling 3.** Each query computes the percentile over `fct_delivery` **in** `POS_Analytics`, not over the raw `SdDocumentLog` events. That table is 15 minutes fresh with the first-event rule already applied, so the replica never sees an interactive first-event scan.
- It is still exact INC per query, and it is still capped at 92 days, as the owner ruled.
- The cap can be lifted later, on a measured timing (the dbt spike below).
- **Every mart carries `_loaded_at`.** A `mart_load_log` table holds one row per run, with each source's newest event time. Freshness is shown as the screen's **"data as of HH:MM"** (755): the newest event time from the last good run, not the moment the run finished.
- **The dimensions** (store, city, region, channel map, the Sunday week, dates capped at today) are dbt seeds and models in the same project. Cube declares them over the marts once. The `DocumentSource` → five-group map is a dbt seed, and it is the only copy.
- `Material` and `StoreGroup` being missing does not matter for v1: there is no product dimension, and region comes from `StoreArea`.

Refresh cadence. There are two dbt jobs per area, selected by tag:

- **`intraday`, every 15 minutes:**
- It rebuilds today's slice, and before 02:00 yesterday's too.
- It builds the slice into a staging table, then swaps it in inside one transaction. A failed run leaves the last good slice in place; it never leaves a gap.
- **`nightly`, at 03:00 (in 750's off-hours window):**
- It rebuilds the last three days, to pick up late till uploads and out-of-order log events.
- It runs `dbt test`: uniqueness, not-null, and the till-value reconciliation of the daily mart against a direct `RetailTrx` sum for yesterday.
- It runs `dbt source freshness`.
- **Bounds:** dbt runs with one or two threads, and every source query is bounded by date. The replica only ever sees bounded, scheduled `SELECT`s from the dbt login. It sees none of Cube's interactive traffic.
- **Area 4** joins the same two jobs once 753 lands.

Who owns and runs the jobs.

- **We (the BackOffice dev team) own the dbt project.**

- It lives in the BackOffice repo beside the Cube model: one `Analytics/` folder with `dbt/` and `cube/`. The definitions sit in one reviewable tree.
- That partly clears the "who keeps the semantic model" fog. The review process stays fog.
- **The scheduler runs on the analytics host** that 749 provisions (Linux or Docker, beside Cube): a scheduled container running `dbt build --select tag:…`.
- It does **not** run in SQL Agent on `DWPOSREPL03`. That keeps Python off the DB server and keeps our code out of the DBA's job list.
- **The DBA (the owner, acting as DBA, as in 753) owns:**
- the database `POS_Analytics`, its size budget and its backups (or none, since it can be rebuilt);
- one writer login, provisionally **`svc_analytics_dbt`**: `db_owner` on `POS_Analytics`, `db_datareader` on `POS`;
- Cube's reader login, provisionally **`svc_analytics_cube`**: `db_datareader` on `POS_Analytics` **only**.
- With Enterprise edition available, the DBA is also *asked* (not required) for a Resource Governor workload group capping both logins (750 §4, item 1).

When the replica is reinitialised:

- **`POS_Analytics` survives.** Reinitialisation touches only the subscription database's articles (742 §2a). The marts and `fct_*` keep their history, and the screens keep answering, with "data as of" visibly falling behind.
- **During the reinit**, the source-freshness check fails, so runs fail **closed**: no swap, and the last good slice stays.
- The `mart_load_log` row records the failure.
- The board shows the stale "as of"; it does not show an empty chart.
- **After the reinit**, a runbook step, `dbt build --vars '{lookback_days: N}'`, rebuilds the days the outage spanned. It is **never** a `--full-refresh` of the history: a year of `RetailTrxDetail`-scale extraction took about 20 minutes on joins (742).
- **If the reinit dropped the subscriber-side NCCI columnstores** (750 §4, item 3), the users do not notice, because they read marts. Only the dbt runs slow down. That makes DBA item 3 a question of **run duration**: whether a 15-minute intraday run still fits inside 15 minutes on rowstore. It no longer decides what users see.

734's `sales_summary` rollup.

- **`mart_till_sales_daily` is that rollup. There is one permanent copy.**
- BackOffice 3346 (spec 3345, for the pitch) builds its own nightly-plus-15-minute rollup on HQ "if none exists". This map is separate from the pitch, and `POS_Analytics` will not exist by about 2026-12-03, so **3346 proceeds as specified.**
- At parity, 746 already re-points `sales_summary` to `semantic_query`. That same step **retires 3346's rollup and its job**. Until then there are two copies, deliberately, and the parity check is what ends that.
- This is a **BackOffice ask for the spec hand-off**, not filed from here (BackOffice is read-only from this map). It has two parts:
- 3346 records where its rollup lives;
- the parity ticket deletes it.

Re-open triggers (from 750, still owed)

750 item	If it comes back…	Then
9: the sibling DB is refused	No `POS_Analytics` on `DWPOSREPL03`	**Fallback:** the same dbt project targets a separate on-prem SQL Server, pulling over the `read` login by bounded query (742 option c). If no host is allowed either: option 1, with Cube Store rollups and percentiles computed live on the replica

750 item	If it comes back…	Then
7: the four-parallel test moves replication lag	The replica can't carry dbt runs and interactive queries together	Resource Governor workload group, `threads: 1`, and widen intraday to 30 minutes. The owner rules on the cadence change
The off-hours run is much slower than the busy-hour run	Something else heavy runs at night (750 §4, item 4)	Move `nightly` to the quiet hour the DBA names
3: columnstores aren't re-created after a reinit	Intraday runs overrun 15 minutes on rowstore	The DBA scripts NCCI re-creation into the reinit runbook, or intraday widens

What this hands on

- **757 (new):** a local spike proving `dbt-sqlserver` builds the incremental marts, and how long `fct_delivery` and the percentile take. The ADR should not name a tool nobody has run (the 756 rule).
- **749:** the checklist gains `POS_Analytics`, the two `svc_analytics_*` logins and the dbt scheduler container on the analytics host. Its path list gains *analytics host* → `DWPOSREPL03` *:1433*.
- **756:** Cube reads mart tables, not raw till tables. Its rollup step (item 5) stays as a check that nothing is written to the source DB, but it is no longer the v1 serving path.
- **The map's fog:** the freshness patch is cleared. The "guarding cost and load" patch narrows to Cube's result cache and the replica's concurrency, which is sized by 750.

type	wayfinder-ticket
wayfinder	research
map	741
status	done
blocked-by	—

742 — What analytical load can the replica carry, and where may derived tables live?

Question

Can the production read replica (`172.23.27.47` / `POS`) serve **interactive** self-service queries directly? Where may the marts, rollups and semantic-layer caches we derive from it live?

The answer should cover: - **The tables in scope** for the likely subject areas: - till sales: `RetailTrx`, `RetailTrxDetail`; - orders: `SdDocumentHeader` and its document source, plus the delivery document action log; - dispatch and callouts; - store master data. For each, give the row count, which indexes exist (columnstore or not) and the typical latency of representative aggregates: one day, one week and 92 days, by store, city and channel. - **Headroom:** - what else hits the replica (reports, ticket 1260's door, BI extracts); - its hardware and edition, if the DMVs show them; - replication latency; - whether a burst of concurrent analytical queries would hurt replication. - **Where derived objects may live:** 1. a sibling database on the same SQL instance; 2. views or indexed views inside `POS`; 3. a separate on-prem server (SQL Server, or a columnar store such as ClickHouse or DuckDB). Say which needs a DBA and which a replication change. A subscriber's schema can be wiped on reinitialisation. - **What we lack:** missing tables or columns that would block a subject area.

Read-only from the corporate network, through the sanctioned `sqlcmd` shape. If the network is absent, hand the queries to the owner. List the questions only a DBA can answer as a checklist.

Answer

Half answered (2026-11-03). The replica was unreachable from this machine (Parallels NAT `10.211.55.3`, TCP 1433 closed). The documentation half is settled. The measurements are handed to the owner as The owner runs the replica headroom probe and the DBA answers its checklist. Full note: 741-replica-headroom-and-derived-tables.md. Query pack: 741-replica-probe.sql.

- **Nothing derived lives inside `POS`.**
- The `read` login cannot create anything there.
- An indexed (schema-bound) view on a replicated table would block the `drop`-mode re-creation on the next reinitialisation.
- Plain views add nothing a semantic layer doesn't already give.
- **Two viable homes remain:**
- a **sibling DB on `DWPOSREPL03`**: needs a DBA, no replication change, shares the hardware;
- a **separate on-prem server**: needs an IT host, reads through the existing `read` login, isolates load.
- The headroom numbers decide between them, in 748.
- **Direct interactive query is bounded.**
- The columnstore tables (`RetailTrx`, `RetailTrxDetail`) can serve days to weeks directly.

- A full year takes 1–2 minutes, and the shipped audit reports already need a 1,800 s time-out on this box.
- Orders (`SdDocumentHeader`) and the action log (`SdDocumentLog`) have no recorded columnstore.
- So 92-day-and-up dashboards need pre-aggregates somewhere.
- **The edition is unknown and pivotal.** Resource Governor, the only server-side cap, is Enterprise only in SQL Server 2019. On Standard, every guard lives in the semantic layer.
- **Gaps that block a subject area:**
- `Callout*` / `DispatchBoardRelease` are probably not replicated (dispatch needs new articles, which is a DBA job);
- `LoyCustomerFirstPurchase*` is absent.
- **A side-risk:** the replica's NCCI columnstores were added on the subscriber, so a `drop`-mode reinitialisation would silently remove them. This is DBA checklist item 3.

type	wayfinder-ticket
wayfinder	task
map	741
status	claimed
blocked-by	—

750 — The owner runs the replica headroom probe and the DBA answers its checklist

Question

Not a decision. These are the measurements that What analytical load can the replica carry, and where may derived tables live? could not take, because the replica answers only on the corporate network. **HITL:** the owner runs it, or an agent session does on the corporate network. The DBA answers the questions only a DBA can.

1. On the corporate network (`Test-NetConnection 172.23.27.47 -Port 1433` → `True`), run 741-replica-probe.sql **twice**: - once at a busy hour, to `docs/research/741-replica-probe.out.txt`; - once off-hours, to `…offhours.txt`. The command is in the file header. It is read-only and writes nothing.
2. Put the DBA checklist (§4 of 741-replica-headroom-and-derived-tables.md) to the DBA. Record the answers.
3. Fold both into that note's §3–§5. Read P6 against the thresholds in §3.

Resolved when the note records: - the edition; - which in-scope tables are missing; - the P6 timings (busy and off-hours); - the replica lag; - the DBA's answers on items 3 (columnstores on reinit), 5–7 (latency and concurrency), 8 (missing articles) and 9 (sibling DB).

Progress (2026-11-03, not yet resolved)

Step 1 is half done. The busy-hour run took place on Tuesday 2026-11-03 at 20:36 from the corporate network. Its output is 741-replica-probe.out.txt. It is folded into §1, §3a, §4 and §5 of the note.

Resolution criteria now recorded: - **Edition:** Enterprise. Resource Governor is available; whether it is configured is still a DBA question. - **Missing tables:** `Callout`, `CalloutBid`, `CalloutEvent`, `DispatchBoardRelease`, `Material`, `StoreGroup`, `DocumentSource` and `LoyCustomerFirstPurchase`. - **P6, busy hour:** every 92-day aggregate takes 3.3 s or less. That is the first band: direct query through a cached semantic layer. - **Replica lag:** seconds. The 1,236-min `RetailTrx` figure is a date-only-column artifact. - **DBA item 6:** answered by the probe. RCSI is on.

Still owed, before `done`: - [ ] **The off-hours run** (around 03:00): the same command with `-o docs\research\741-replica-probe.offhours.txt`. Fold it into §3a. - [ ] **The DBA's answers** on §4, items 3 (columnstores on reinit), 5 (latency), 7 (the four-parallel test with Replication Monitor), 8 (the eight missing articles: cost and willingness; the v1 dispatch subset is now its own task, 753, and `DispatchBoardRelease` is not a table) and 9 (the sibling DB; `ZC` and `BusinessMonitor` are a precedent). Items 1 (Resource Governor configured?), 2, 4 and 10 are wanted too, but they do not gate this ticket.

741 / 742 — Replica headroom, and where derived tables may live

Research for What analytical load can the replica carry, and where may derived tables live?. Written 2026-11-03. The first draft could not reach the replica (`172.23.27.47` / `POS`, `DWPOSREPL03`): `Test-NetConnection … -Port 1433` returned `TcpTestSucceeded = False` from the Parallels NAT address `10.211.55.3`. So: - §1–§2 come from what the estate has already recorded (BackOffice ticket 1260, BackOffice memory, the BackOffice code) plus Microsoft documentation. - §3 is the read-only query pack, 741-replica-probe.sql.

Updated 2026-11-03 20:36 (ticket 750). The probe was run at the **busy hour** from the corporate network. The output is 741-replica-probe.out.txt. The results are in **§3a**, and §1, §4 and §5 are corrected to match. **Still owed:** - the off-hours run (`741-replica-probe.offhours.txt`); - the DBA's answers (§4).

`[n]` refers to **Sources** (§6).

---

1. Summary

- **The replica is SQL Server 2019** (RTM-CU32-GDR) on `DWPOSREPL03`, a separate host from the live OMS (`DWPOS03`, `172.23.27.50`, off limits) [1]. The `read` login is `db_datareader`, with `CREATE TABLE = 0` [1]. **Nothing can be created in `POS` by us.** Every placement option below needs a DBA.
- **The edition is Enterprise** (measured, §3a: `Enterprise Edition (64-bit)`, EngineEdition 3, 15.0.4470.1). So **Resource Governor is available** [6] and can cap analytics CPU and memory per workload. Whether it is *configured* is a DBA question (§4, item 1).
- **Till sales are analytics-ready. Orders are fast enough at v1 windows, despite having no columnstore.**
- `RetailTrx` (465M rows), `RetailTrxDetail` (1.006B) and `RetailTrxPayment` (501M) carry NCCI columnstores (measured).
- `SdDocumentHeader` (16.8M), `SdDocumentLine` (36.6M) and `SdDocumentLog` (87.9M) have only a clustered PK. There are no NC indexes and no index leads on a date [3].
- Even so, a 92-day aggregate takes about 1.7 s on orders and about 3.2 s on the action log at the busy hour (§3a).
- Earlier figures still stand for longer windows: a full-year aggregate takes 1–2 minutes, and a year-to-date extract with joins took about 20 minutes [2].
- **The heavy-report precedent is already on this box.** The SAP audit reports run their `RetailTrx` date-range aggregates against this replica, with a **1,800 s** command time-out. That was widened from 600 s because store-group aggregates "were throwing Execution Timeout Expired on a busy quarter" [1].
- **Measured (§3a): every 92-day aggregate in the pack returns in under 3.3 s** at a busy hour, with a single session. So direct query through a cached semantic layer is enough for v1 up to about a quarter.
- **A full year (1–2 minutes) is still not interactive.** Windows beyond about 92 days need pre-aggregates, or a cap in the semantic model.
- Concurrency is not yet measured (§4, item 7).
- **Placement, in brief** (detail in §2): **never inside `POS`**. Derived tables go in either a **sibling database on the same instance** (cheapest, but shares the hardware) or a **separate on-prem server** (isolates load, costs a host and a copy lag). Which one depends on the headroom numbers. That choice is the job of What builds the marts.

- **Callouts are missing from the replica (confirmed, §3a).** The `Callout*` tables are recent additions to the OMS app database (spec 2952, 2026-10). A table appears on a transactional subscriber only once someone adds it as an article to the publication [4].
- `Callout`, `CalloutBid`, `CalloutEvent` and `DispatchBoardRelease` are all absent.
- So are `Material`, `StoreGroup`, `DocumentSource` and `LoyCustomerFirstPurchase`.
- The dispatch subject area therefore needs a **replication change**, which is a DBA job.

2. Where derived objects may live

	(a) Sibling DB on the same instance	(b) Views or indexed views inside `POS`	(c) Separate on-prem server
Example	`POS_Analytics` beside `POS` on `DWPOSREPL03`	`dbo.v_SalesDay` in `POS`	a new SQL Server, ClickHouse or DuckDB host
Survives reinitialisation	**Yes.** Reinit touches only the subscription DB's articles	Plain views: they stay in the catalogue. **Indexed (schema-bound) views: they break it** (below)	Yes
Needs a DBA	Yes: `CREATE DATABASE`, a writer login on it, a SQL Agent job for refresh	Yes: DDL rights in `POS`, which the `read` login refuses [1]	To read: no, the existing `read` login suffices. To host: IT (ticket 749)
Needs a replication change	No	No (plain) / effectively yes (indexed: `@pre_creation_cmd` must stop being `drop`)	No, if it pulls by query. Yes, if it becomes a second subscriber
Load isolation	**None.** It shares CPU, memory, tempdb and I/O with replication apply	None	**Full.** The replica sees only scheduled, bounded extracts
Freshness	Refresh cadence (for example every 15 minutes or nightly) on top of replica lag	Live (plain views)	Extract cadence on top of replica lag
Cross-DB constraints	Indexed views cannot span databases, so rollups are tables refreshed by a job [5]	—	A copy pipeline to build and run (incremental on `TrxDate` / `EntryTime`)
Verdict	**Viable.** Default if headroom is ample	**Ruled out**	**Viable.** Default if headroom is thin, or on Standard edition

Why (b) is ruled out: - A subscriber article's table is normally re-created on reinitialisation (`@pre_creation_cmd = 'drop'`, the default) [4]. - A table that a `SCHEMABINDING` view references **cannot be dropped** while that view exists [5]. So an indexed view on a replicated table would make the next snapshot fail. The replication doc does not say this in so many words; it is an inference from [4] and [5]. - Plain views would survive, but they add nothing a semantic layer doesn't already give. They also mean asking a DBA for DDL rights in a database we must treat as disposable.

A side-risk the same fact exposes: the replica's NCCI columnstores are not publisher indexes. They were added on the subscriber, so a `drop`-mode reinitialisation would **silently remove them**. The 1–2 minute full-year figure would then become a 1B-row rowstore scan. That puts it on the DBA checklist (§4, item 3).

3. What is still unmeasured — the probe pack

741-replica-probe.sql is read-only and date-bounded. It writes nothing (not even `#temp`) and runs under the `read` login. It covers each item the ticket asked for:

Probe	Answers
P1	Edition, engine edition, version. RCSI / snapshot isolation flags (readers blocking the distribution agent). Query Store. Any sibling DB already present
P2	Which in-scope tables are **present or missing** (`Callout*`, `DispatchBoardRelease`, `LoyCustomerFirstPurchase`, …). Row counts. Every index and its type. Key columns on the four big tables. Count of schema-bound objects already in `POS`
P3	`MSreplication_subscriptions` last-agent-write time. Newest-row lag on `RetailTrx`, `SdDocumentHeader`, `SdDocumentLog`
P4	Whether `read` holds `VIEW SERVER STATE` / `VIEW DATABASE STATE` (expected: no. The OMS login is refused it [7])
P5	If allowed: CPUs, memory, sessions by login and program (*who else hits it*), top waits. If not: deferred to the DBA
P6	Timed aggregates: 1 day, 7 days and 92 days, by store, city and channel, on `RetailTrx` (header), `RetailTrxDetail` (lines), `SdDocumentHeader` (by store × `DocumentSource`, by city × `DocumentType`), `SdDocumentLog` (by store × `ActionType`), and `Callout` if present

It runs twice: once at a busy hour and once off-hours. Columns used were checked against the BackOffice models (`RetailTrx.cs`, `SdDocumentHeader.cs`, `SdDocumentLog.cs`, `Store.cs`): `Store` carries `City`, `Area` and `Country`. Orders have `DocumentSource` and `DocumentType`; tills have `DocumentType`.

How to read P6 once it is back (the thresholds 748 will use): - 92-day by-store answers in **under about 5 s** on all four tables: direct query through a cached semantic layer is enough for v1. Derived tables are optional. - Columnstore tables are fast but **orders and the log scan for tens of seconds**: pre-aggregate those two only (sibling DB or separate server). - Anything **over about 60 s at a busy hour**, or replication lag that grows while P6 runs: separate server, extracts only, and the replica never sees interactive traffic.

3a. Measured — busy hour, Tuesday 2026-11-03 20:36 (ticket 750)

How it was run: - The `read` login ran the pack through the sanctioned `sqlcmd` shape. It took 29 s end to end. The raw output is 741-replica-probe.out.txt. - Two small read-only follow-ups were run (named below). They were not saved; their results are recorded here. - **The off-hours run is still owed.**

Probe	Result
P1 edition	**Enterprise Edition (64-bit)**, EngineEdition 3, 15.0.4470.1
P1 `POS` flags	compat 150, SIMPLE recovery, **`is_read_committed_snapshot_on = 1` (RCSI on)**, snapshot isolation OFF, **Query Store OFF**
P1 other user DBs	**`ZC`** (created 2025-09-30) and **`BusinessMonitor`** (2025-10-01) already sit beside `POS`. So the instance already hosts sibling DBs: a precedent for option (a)
P2 missing (as `dbo.`)	`Callout`, `CalloutBid`, `CalloutEvent`, `DispatchBoardRelease`, `DocumentSource`, `LoyCustomerFirstPurchase`, `Material`, `StoreGroup`. The follow-up found no other-schema twin. The only near names are `MaterialDimension` and `MaterialTaxClassification`. `SdDocumentSource` (43 rows) is present and is the order-source lookup
P2 present	`RetailTrx`, `RetailTrxDetail`, `RetailTrxPayment`, `SdDocumentHeader`, `SdDocumentLine`, `SdDocumentLog`, `SdDocumentSource`, `SdDocumentActionType`, `SdDocumentHeaderStatus`, `SdDocumentDeliveryType`, `SdDocumentType`, `Staff`, `Store` (1,561)

Probe	Result
P2 indexes	NCCI on `RetailTrx` (465,331,893 rows), `RetailTrxDetail` (1,005,935,404) and `RetailTrxPayment` (501,373,182). The order tables have the clustered PK only: `SdDocumentHeader` (16.8M, `DocumentNo`), `SdDocumentLine` (36.6M), `SdDocumentLog` (87.9M, `LogNo`). Till PKs lead on `Client, StoreCode, MachineCode, TrxNumber`. **No rowstore index leads on a date**
P2 schema-bound objects	34 views, of which **one is schema-bound: `dbo.Wsp_PostPurchaseSurvey`**. The follow-up shows it has **no index** and no visible table reference. It is not the indexed-view reinit hazard of §2, but the DBA should know it is there
P3 replication	20 subscriptions, all writing 10 s before the probe: 18 from `DWPOS03/POS`, plus `NPDB01/Nphies` (2) and `DWWASDB01/Wasfaty` (1). **`POS` here is a merge point of three publishers**, not a plain copy of the OMS database
P3 lag	`SdDocumentHeader` and `SdDocumentLog`: newest row 0 min old. `RetailTrx` reported 1,236 min, but that is an **artifact**: `TrxDate` is date-only, 00:00. The follow-up on `MAX(TimeWhenTrxClosed)` today gave **about 2 s** (156,551 till transactions by 20:37). **End-to-end lag at the busy hour: seconds**
P4 grants	`VIEW SERVER STATE` 0, `VIEW DATABASE STATE` 0, `SHOWPLAN` 0
P5	Denied. Hardware and other traffic go to the DBA (§4, items 2 and 4)

P6 timings (ms, single session, busy hour):

Query	ms	rows
RetailTrx 1d by store	313	955
RetailTrx 7d by store	375	963
RetailTrx 92d by store	1,078	972
RetailTrx 92d by city	954	203
RetailTrx 92d by channel × day	1,157	1,107
RetailTrxDetail 7d by store	907	965
RetailTrxDetail 92d by store	1,813	974
SdDocumentHeader 1d by store × source	735	6,777
SdDocumentHeader 7d by store × source	1,532	12,343
SdDocumentHeader 92d by store × source	1,719	17,591
SdDocumentHeader 92d by city × type	1,517	1,820
SdDocumentLog 7d by store × action	2,908	5,476
SdDocumentLog 92d by store × action	3,220	9,878
Callout 92d	—	table missing

Read against the §3 thresholds: - The **first band** applies. All four tables answer a 92-day by-store aggregate in **under 5 s** (the worst is 3.2 s), so **direct query through a cached semantic layer is enough for v1**, and derived tables are optional at windows of up to about a quarter. - **Caveats:** - This is one session, in one busy hour, and the cache may be warm. The off-hours run and the DBA's four-parallel test (§4, item 7) decide whether the band holds under concurrency. - Windows beyond 92 days still meet the 1–2 minute full-year figure [2]. - The orders and the log scan their clustered PK. Their time grows with the window, not the filter.

4. DBA checklist — what only a DBA can answer

Status after §3a: - **The probe answered items 1 (edition: Enterprise) and 6 (RCSI on).** Item 1's "is Resource Governor configured?" is still open. - Items 2, 3, 4, 5, 7, 8, 9 and 10 are **still owed to the DBA**. - Item 8's list is now exact. Item 9 has a precedent: `ZC` and `BusinessMonitor` already sit on the instance.

1. **Edition** of `DWPOSREPL03`, if P1 is ambiguous, and whether Resource Governor is configured.
2. **Hardware:** cores, memory, storage. What `max server memory` and MAXDOP are set to.
3. **The NCCI columnstores on `RetailTrx` / `RetailTrxDetail`:** - Who created them on the subscriber? - Are they scripted to be re-created after a reinitialisation (a post-snapshot script, or a runbook)? - What is each article's `@pre_creation_cmd`?
4. **Who else hits the replica** besides the SAP audit reports and ticket 1260's door: other reports, BI extracts, SSRS, ad-hoc users. What are its busy hours?
5. **Replication latency:** the typical and worst-case latency (tracer tokens from the publisher, or Replication Monitor), and whether it has ever fallen behind during heavy reports.
6. **Isolation:** is RCSI on for `POS` on the subscriber? If not, will the DBA turn it on, so analytics readers cannot block the distribution agent?
7. **A concurrency test the DBA watches:** four parallel 92-day P6 queries, with Replication Monitor open. Does latency move?
8. **Missing articles:** would the DBA add `Callout`, `CalloutBid`, `CalloutEvent` and `DispatchBoardRelease` (and any table P2 lists as missing) to the publication? What does adding an article cost (a snapshot of only the new articles, or a reinitialisation)?
9. **A sibling database:** may `POS_Analytics` (or a similar name) be created on `DWPOSREPL03`? That includes a writer service login, a SQL Agent refresh job, and a size budget.
10. **Grant:** may the `read` login (or a new analytics login) get `VIEW DATABASE STATE` on `POS`, so the semantic layer can read Query Store or index usage for its own tuning?

5. What we lack (blocks a subject area)

Gap	Blocks	Fix
`Callout*`, `DispatchBoardRelease`: **not articles (measured)**	Dispatch and callouts	Add the articles (DBA, publisher side)
`Material`, `StoreGroup`: **not on the replica (measured)**. Only `MaterialDimension` and `MaterialTaxClassification` are there	Any by-product or by-category cut, and any by-store-group cut	Add the articles, or source the masters elsewhere into the mart (DBA, or decided in 748)
`LoyCustomerFirstPurchase*`: not on the replica [3]	First-purchase / new-customer counts	Add the article, or derive it from `RetailTrx` in the mart
No publisher NC indexes on the replica [3]	Fast order and log aggregates	A derived table, or a subscriber-side index the DBA owns (with the same reinit caveat as §2)
`Staff.StaffClass` blank estate-wide [2]	Any by-role cut (for example "pharmacist")	Out of v1 unless master data is fixed
Placeholder loyalty IDs (`3795407`, `3378`, `1000000004`) [2]	Per-member measures (none are planned: no PII)	Exclude them in the semantic model if a member *count* is ever offered
Till clocks dated 2028/2029 [2]	Any open-ended "to date" window	The semantic model caps every window at today

6. Sources

1. BackOffice ticket 1260, *A read-only reach into production*, `## Answer` (2026-08-24): server identity, version, `read` role and object grants, the `AuditConnections.QueryTimeoutSeconds = 1800` precedent. `C:\Work\DMSCO\BackOffice\.issues\1260-read-only-reach-into-production.md`.
2. BackOffice memory `reference_replica_retailtrxdetail_sales_analysis` (2026-10-22).
3. BackOffice memory `reference_pos_replica_readonly_door` (2026-09-10).
4. Microsoft Learn, sys.sp addarticle (Transact-SQL): `@pre_creation_cmd` = none / drop / delete / truncate. Adding articles needs a snapshot or a reinitialisation, depending on the change.
5. Microsoft Learn, CREATE VIEW (Transact-SQL): tables in a `SCHEMABINDING` view cannot be dropped while it exists. Schema-bound views reference only objects in the same database.
6. Microsoft Learn, Editions and supported features of SQL Server 2019 and Resource Governor: Resource Governor is Enterprise/Developer only before SQL Server 2025.
7. BackOffice `.issues/assets/3029-probe-4-log-and-locks.sql`, header comment: "the OMS login is refused" `VIEW SERVER STATE`.

Source: oms-react/docs/research/741-replica-probe.sql

```
/* ================================================================================================
   741 / ticket 742 — replica headroom probe. READ-ONLY.

   Target : 172.23.27.47 / POS (DWPOSREPL03), the `read` login (db_datareader, CREATE TABLE = 0).
   Writes : none. No #temp, no table variables over data, no indexes. Every data read is date-bounded.
   Run    : on the corporate network, in SSMS, or with the sanctioned sqlcmd shape plus `-i` and `-o`:
               $b = New-Object System.Data.SqlClient.SqlConnectionStringBuilder `
                    ([Environment]::GetEnvironmentVariable('POS_REPLICA_READONLY','User'))
               sqlcmd -S $b.DataSource -d $b.InitialCatalog -U $b.UserID -P $b.Password -C -l 20 `
                      -t 1800 -W -s "|" -i docs\research\741-replica-probe.sql `
                      -o docs\research\741-replica-probe.out.txt
            Run it TWICE: once at a busy hour (about 13:00 or 21:00 local) and once off-hours. Rename
            the second output to *.offhours.txt. Paste both back, or leave them beside this file
            (they hold aggregates only, no personal data).
   Time   : P6 dominates. Expect 5–20 minutes in total. Every section prints its own elapsed ms.
   ================================================================================================ */
SET NOCOUNT ON;
SET ANSI_WARNINGS OFF;   -- suppresses "Null value is eliminated" noise in the output
DECLARE @t0 DATETIME2, @label VARCHAR(80);

/* P1 — identity, edition, database flags ------------------------------------------------------- */
SELECT  [Server]        = @@SERVERNAME,
        [Edition]       = SERVERPROPERTY('Edition'),
        [EngineEdition] = SERVERPROPERTY('EngineEdition'),   -- 3 = Enterprise/Developer, 2 = Standard
        [Version]       = SERVERPROPERTY('ProductVersion'),
        [Now]           = SYSDATETIME();

SELECT  name, compatibility_level, recovery_model_desc, is_subscribed, is_published,
        is_read_committed_snapshot_on, snapshot_isolation_state_desc,
        is_query_store_on, create_date
FROM    sys.databases
WHERE   database_id > 4;                                        -- every user DB: shows any sibling

/* P2 — the in-scope tables: present?, rows, every index ---------------------------------------- */
SELECT  t.TableName,
        Present = CASE WHEN OBJECT_ID('dbo.' + t.TableName, 'U') IS NULL THEN 'MISSING' ELSE 'yes' END
FROM (VALUES ('RetailTrx'),('RetailTrxDetail'),('RetailTrxPayment'),
             ('SdDocumentHeader'),('SdDocumentLine'),('SdDocumentLog'),('SdDocumentSource'),
             ('DocumentSource'),('SdDocumentActionType'),('SdDocumentHeaderStatus'),
             ('SdDocumentDeliveryType'),('SdDocumentType'),
             ('Callout'),('CalloutBid'),('CalloutEvent'),('DispatchBoardRelease'),
             ('Store'),('StoreGroup'),('Staff'),('Material'),
             ('LoyCustomerFirstPurchase'),('MSreplication_subscriptions')) AS t(TableName)
ORDER BY Present, t.TableName;

SELECT  TableName  = o.name,
        IndexName  = ISNULL(i.name, '(heap)'),
        IndexType  = i.type_desc,                               -- look for NONCLUSTERED COLUMNSTORE
        i.is_primary_key, i.has_filter,
        [Rows]     = SUM(CASE WHEN i.index_id IN (0,1) OR i.type IN (5,6) THEN p.rows ELSE 0 END),
        Partitions = COUNT(*)
FROM    sys.objects o
JOIN    sys.indexes i    ON i.object_id = o.object_id
JOIN    sys.partitions p ON p.object_id = i.object_id AND p.index_id = i.index_id
WHERE   o.type = 'U'
  AND   o.name IN ('RetailTrx','RetailTrxDetail','RetailTrxPayment','SdDocumentHeader','SdDocumentLine',
                   'SdDocumentLog','SdDocumentSource','DocumentSource','Callout','CalloutBid',
                   'CalloutEvent','DispatchBoardRelease','Store','Material')
GROUP BY o.name, i.name, i.type_desc, i.is_primary_key, i.has_filter, i.index_id
ORDER BY o.name, i.index_id;
```

```
-- Key columns of every index on the four big tables (does any rowstore index lead on a date?)
SELECT  TableName = OBJECT_NAME(ic.object_id), IndexName = i.name, ic.key_ordinal,
        ColumnName = c.name, ic.is_included_column
FROM    sys.index_columns ic
JOIN    sys.indexes i ON i.object_id = ic.object_id AND i.index_id = ic.index_id
JOIN    sys.columns c ON c.object_id = ic.object_id AND c.column_id = ic.column_id
WHERE   OBJECT_NAME(ic.object_id) IN ('RetailTrx','RetailTrxDetail','SdDocumentHeader','SdDocumentLog')
  AND   i.type <> 6                                          -- NCCI lists every column: skip
ORDER BY TableName, IndexName, ic.is_included_column, ic.key_ordinal;

-- Who else lives in POS: user views / procs / schema-bound views (reinit hazards)
SELECT  o.type_desc, Cnt = COUNT(*),
        SchemaBound = SUM(CASE WHEN OBJECTPROPERTY(o.object_id, 'IsSchemaBound') = 1 THEN 1 ELSE 0 END)
FROM    sys.objects o WHERE o.type IN ('V','P','FN','IF','TF') AND o.is_ms_shipped = 0
GROUP BY o.type_desc;

/* P3 — replication state and lag proxies ------------------------------------------------------- */
IF OBJECT_ID('dbo.MSreplication_subscriptions') IS NOT NULL
    EXEC('SELECT publisher, publisher_db, publication, subscription_type, [time] AS LastAgentWrite,
                DATEDIFF(SECOND, [time], GETDATE()) AS SecondsSinceAgentWrite
          FROM dbo.MSreplication_subscriptions');
ELSE SELECT 'no MSreplication_subscriptions visible to this login' AS ReplicationState;

-- Freshness of the data itself: newest row vs. now (a crude end-to-end lag figure)
SELECT  Src = 'RetailTrx',        Newest = MAX(TrxDate),   LagMin = DATEDIFF(MINUTE, MAX(TrxDate), GETDATE())
FROM    RetailTrx        WHERE TrxDate >= DATEADD(DAY, -2, GETDATE()) AND TrxDate <= GETDATE()
UNION ALL
SELECT  'SdDocumentLog',          MAX(EntryTime),          DATEDIFF(MINUTE, MAX(EntryTime), GETDATE())
FROM    SdDocumentLog    WHERE EntryTime >= DATEADD(DAY, -2, GETDATE()) AND EntryTime <= GETDATE()
UNION ALL
SELECT  'SdDocumentHeader',       MAX(EntryTime),          DATEDIFF(MINUTE, MAX(EntryTime), GETDATE())
FROM    SdDocumentHeader WHERE EntryTime >= DATEADD(DAY, -2, GETDATE()) AND EntryTime <= GETDATE();

/* P4 — what this login may see about load (decides which P5 rows come back) ---------------------- */
SELECT  ViewServerState   = HAS_PERMS_BY_NAME(NULL, NULL, 'VIEW SERVER STATE'),
        ViewDatabaseState = HAS_PERMS_BY_NAME(DB_NAME(), 'DATABASE', 'VIEW DATABASE STATE'),
        ShowPlan          = HAS_PERMS_BY_NAME(DB_NAME(), 'DATABASE', 'SHOWPLAN');

/* P5 — hardware and other traffic (only if VIEW SERVER STATE; otherwise a DBA question) ---------- */
IF HAS_PERMS_BY_NAME(NULL, NULL, 'VIEW SERVER STATE') = 1
BEGIN
    EXEC('SELECT cpu_count, hyperthread_ratio, physical_memory_kb / 1024 AS MemoryMB,
                committed_target_kb / 1024 AS BufferTargetMB, sqlserver_start_time
          FROM sys.dm_os_sys_info');
    EXEC('SELECT login_name, program_name, Sessions = COUNT(*),
                CpuMs = SUM(cpu_time), Reads = SUM(logical_reads)
          FROM sys.dm_exec_sessions WHERE is_user_process = 1
          GROUP BY login_name, program_name ORDER BY CpuMs DESC');
    EXEC('SELECT TOP 10 wait_type, wait_time_ms, waiting_tasks_count FROM sys.dm_os_wait_stats
          WHERE wait_type NOT LIKE ''SLEEP%'' AND wait_type NOT LIKE ''%QUEUE%''
            AND wait_type NOT IN (''BROKER_TASK_STOP'',''CHECKPOINT_QUEUE'',''XE_TIMER_EVENT'',

''DIRTY_PAGE_POLL'',''HADR_FILESTREAM_IOMGR_IOCOMPLETION'',''SOS_WORK_DISPATCHER'')
          ORDER BY wait_time_ms DESC');
END
ELSE SELECT 'VIEW SERVER STATE denied: hardware + other-traffic rows go to the DBA checklist' AS P5;

/* P6 — representative aggregates, timed. Anchor = yesterday (a whole day). ---------------------- */
DECLARE @d1 DATETIME = CAST(CAST(GETDATE() AS DATE) AS DATETIME);  -- today 00:00 (exclusive end)
DECLARE @w1 DATETIME = DATEADD(DAY, -1,  @d1),
        @w7 DATETIME = DATEADD(DAY, -7,  @d1),
        @w92 DATETIME = DATEADD(DAY, -92, @d1);
DECLARE @r TABLE (Label VARCHAR(80), Ms INT, ResultRows INT);
DECLARE @n INT;
```

```
-- Till sales, header, by store
SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, Trx = COUNT(*), Amt = SUM(Amount) FROM RetailTrx
    WHERE TrxDate >= @w1 AND TrxDate < @d1 AND TrxStatus = 1 GROUP BY StoreCode) x;
INSERT @r VALUES ('RetailTrx 1d by store', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, Trx = COUNT(*), Amt = SUM(Amount) FROM RetailTrx
    WHERE TrxDate >= @w7 AND TrxDate < @d1 AND TrxStatus = 1 GROUP BY StoreCode) x;
INSERT @r VALUES ('RetailTrx 7d by store', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, Trx = COUNT(*), Amt = SUM(Amount) FROM RetailTrx
    WHERE TrxDate >= @w92 AND TrxDate < @d1 AND TrxStatus = 1 GROUP BY StoreCode) x;
INSERT @r VALUES ('RetailTrx 92d by store', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

-- Till sales by city (join to the store master) and by channel (DocumentType), 92 days
SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT s.City, Trx = COUNT(*), Amt = SUM(t.Amount)
    FROM RetailTrx t JOIN Store s ON s.StoreCode = t.StoreCode
    WHERE t.TrxDate >= @w92 AND t.TrxDate < @d1 AND t.TrxStatus = 1 GROUP BY s.City) x;
INSERT @r VALUES ('RetailTrx 92d by city', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT DocumentType, d = CAST(TrxDate AS DATE), Trx = COUNT(*), Amt = SUM(Amount)
    FROM RetailTrx WHERE TrxDate >= @w92 AND TrxDate < @d1 AND TrxStatus = 1
    GROUP BY DocumentType, CAST(TrxDate AS DATE)) x;
INSERT @r VALUES ('RetailTrx 92d by channel x day', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

-- Till lines (1.0B rows): 7d and 92d by store
SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, Lines = COUNT(*), Qty = SUM(Quantity), Amt = SUM(Amount)
    FROM RetailTrxDetail WHERE TrxDate >= @w7 AND TrxDate < @d1 GROUP BY StoreCode) x;
INSERT @r VALUES ('RetailTrxDetail 7d by store', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, Lines = COUNT(*), Qty = SUM(Quantity), Amt = SUM(Amount)
    FROM RetailTrxDetail WHERE TrxDate >= @w92 AND TrxDate < @d1 GROUP BY StoreCode) x;
INSERT @r VALUES ('RetailTrxDetail 92d by store', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

-- Orders (no columnstore expected): 1d, 7d, 92d by store and source
SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, DocumentSource, Docs = COUNT(*), Net = SUM(NetTotal)
    FROM SdDocumentHeader WHERE EntryTime >= @w1 AND EntryTime < @d1 GROUP BY StoreCode, DocumentSource) x;
INSERT @r VALUES ('SdDocumentHeader 1d by store x source', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, DocumentSource, Docs = COUNT(*), Net = SUM(NetTotal)
    FROM SdDocumentHeader WHERE EntryTime >= @w7 AND EntryTime < @d1 GROUP BY StoreCode, DocumentSource) x;
INSERT @r VALUES ('SdDocumentHeader 7d by store x source', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, DocumentSource, Docs = COUNT(*), Net = SUM(NetTotal)
    FROM SdDocumentHeader WHERE EntryTime >= @w92 AND EntryTime < @d1 GROUP BY StoreCode, DocumentSource) x;
INSERT @r VALUES ('SdDocumentHeader 92d by store x source', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT s.City, h.DocumentType, Docs = COUNT(*), Net = SUM(h.NetTotal)
    FROM SdDocumentHeader h JOIN Store s ON s.StoreCode = h.StoreCode
    WHERE h.EntryTime >= @w92 AND h.EntryTime < @d1 GROUP BY s.City, h.DocumentType) x;
INSERT @r VALUES ('SdDocumentHeader 92d by city x type', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

-- Delivery action log: 7d and 92d by store x action
```

```
SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, ActionType, Events = COUNT(*)
    FROM SdDocumentLog WHERE EntryTime >= @w7 AND EntryTime < @d1 GROUP BY StoreCode, ActionType) x;
INSERT @r VALUES ('SdDocumentLog 7d by store x action', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

SET @t0 = SYSDATETIME();
SELECT @n = COUNT(*) FROM (SELECT StoreCode, ActionType, Events = COUNT(*)
    FROM SdDocumentLog WHERE EntryTime >= @w92 AND EntryTime < @d1 GROUP BY StoreCode, ActionType) x;
INSERT @r VALUES ('SdDocumentLog 92d by store x action', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);

-- Callouts, only if replicated
IF OBJECT_ID('dbo.Callout', 'U') IS NOT NULL
BEGIN
    SET @t0 = SYSDATETIME();
    EXEC sp_executesql N'SELECT @n = COUNT(*) FROM (SELECT Status, AwardedParty, Cnt = COUNT(*) FROM Callout
        WHERE OpenedAt >= @f AND OpenedAt < @t GROUP BY Status, AwardedParty) x',
        N'@n INT OUTPUT, @f DATETIME, @t DATETIME', @n OUTPUT, @w92, @d1;
    INSERT @r VALUES ('Callout 92d by status x party', DATEDIFF(MILLISECOND, @t0, SYSDATETIME()), @n);
END

SELECT * FROM @r;
```

Source: oms-react/docs/research/741-replica-probe.out.txt

```
Server|Edition|EngineEdition|Version|Now
------|-------|-------------|-------|---
DWPOSREPL03|Enterprise Edition (64-bit)|3|15.0.4470.1|2026-11-03 20:36:10.4260403
name|compatibility_level|recovery_model_desc|is_subscribed|is_published|is_read_committed_snapshot_on|snapsho
t_isolation_state_desc|is_query_store_on|create_date
----|------------------|------------------|------------|-----------|----------------------------|-------
---------------------|-----------------|-----------
POS|150|SIMPLE|0|0|1|OFF|0|2025-09-27 03:23:05.393
ZC|150|FULL|0|0|0|OFF|0|2025-09-30 11:49:12.903
BusinessMonitor|110|FULL|0|0|0|OFF|0|2025-10-01 09:49:47.620
TableName|Present
---------|-------
Callout|MISSING
CalloutBid|MISSING
CalloutEvent|MISSING
DispatchBoardRelease|MISSING
DocumentSource|MISSING
LoyCustomerFirstPurchase|MISSING
Material|MISSING
StoreGroup|MISSING
MSreplication_subscriptions|yes
RetailTrx|yes
RetailTrxDetail|yes
RetailTrxPayment|yes
SdDocumentActionType|yes
SdDocumentDeliveryType|yes
SdDocumentHeader|yes
SdDocumentHeaderStatus|yes
SdDocumentLine|yes
SdDocumentLog|yes
SdDocumentSource|yes
SdDocumentType|yes
Staff|yes
Store|yes
TableName|IndexName|IndexType|is_primary_key|has_filter|Rows|Partitions
---------|---------|---------|--------------|----------|----|----------
RetailTrx|PK_RetailTrx|CLUSTERED|1|0|465331893|1
RetailTrx|NCCI|NONCLUSTERED COLUMNSTORE|0|0|465331893|1
RetailTrxDetail|PK_RetailTrxDetail|CLUSTERED|1|0|1005935404|1
RetailTrxDetail|NCCI|NONCLUSTERED COLUMNSTORE|0|0|1005935404|1
RetailTrxPayment|PK_RetailTrxPayment|CLUSTERED|1|0|501373182|1
RetailTrxPayment|NCCI|NONCLUSTERED COLUMNSTORE|0|0|501373182|1
SdDocumentHeader|PK_SdDocumentHeader|CLUSTERED|1|0|16790609|1
SdDocumentLine|PK_SdDocumentLine|CLUSTERED|1|0|36626974|1
SdDocumentLog|PK_SdDocumentLog|CLUSTERED|1|0|87871471|1
SdDocumentSource|PK_SdDocumentSource|CLUSTERED|1|0|43|1
Store|PK_Store|CLUSTERED|1|0|1561|1
TableName|IndexName|key_ordinal|ColumnName|is_included_column
---------|---------|-----------|----------|------------------
RetailTrx|PK_RetailTrx|1|Client|0
RetailTrx|PK_RetailTrx|2|StoreCode|0
RetailTrx|PK_RetailTrx|3|MachineCode|0
RetailTrx|PK_RetailTrx|4|TrxNumber|0
RetailTrxDetail|PK_RetailTrxDetail|1|Client|0
RetailTrxDetail|PK_RetailTrxDetail|2|StoreCode|0
RetailTrxDetail|PK_RetailTrxDetail|3|MachineCode|0
RetailTrxDetail|PK_RetailTrxDetail|4|TrxNumber|0
RetailTrxDetail|PK_RetailTrxDetail|5|LineNumber|0
SdDocumentHeader|PK_SdDocumentHeader|1|DocumentNo|0
SdDocumentLog|PK_SdDocumentLog|1|LogNo|0
type_desc|Cnt|SchemaBound
```

```
---------|---|-----------
VIEW|34|1
publisher|publisher_db|publication|subscription_type|LastAgentWrite|SecondsSinceAgentWrite
---------|------------|-----------|-----------------|--------------|----------------------
DWPOS03|POS|Loy|0|2026-11-03 20:36:00|10
DWPOS03|POS|Loy_2|0|2026-11-03 20:36:00|10
DWPOS03|POS|Loy_Family|0|2026-11-03 20:36:00|10
DWPOS03|POS|Master1|0|2026-11-03 20:36:00|10
NPDB01|Nphies|Nphies|0|2026-11-03 20:36:00|10
DWPOS03|POS|OMS|0|2026-11-03 20:36:00|10
DWPOS03|POS|Other1|0|2026-11-03 20:36:00|10
NPDB01|Nphies|Others|0|2026-11-03 20:36:00|10
DWPOS03|POS|RetailTrx|0|2026-11-03 20:36:00|10
DWPOS03|POS|RetailTrxConditions|0|2026-11-03 20:36:00|10
DWPOS03|POS|RetailTrxDetail|0|2026-11-03 20:36:00|10
DWPOS03|POS|RetailTrxDetailPromo|0|2026-11-03 20:36:00|10
DWPOS03|POS|RetailTrxEmail|0|2026-11-03 20:36:00|10
DWPOS03|POS|RetailTrxIndex|0|2026-11-03 20:36:00|10
DWPOS03|POS|RetailTrxPayment|0|2026-11-03 20:36:00|10
DWPOS03|POS|RetailTrxSerial|0|2026-11-03 20:36:00|10
DWPOS03|POS|SapTrx|0|2026-11-03 20:36:00|10
DWPOS03|POS|StockDiscrepancy|0|2026-11-03 20:36:00|10
DWWASDB01|Wasfaty|Wasfaty|0|2026-11-03 20:36:00|10
DWPOS03|POS|Wp|0|2026-11-03 20:36:00|10
Src|Newest|LagMin
---|------|------
SdDocumentLog|2026-11-03 20:36:01.4539861|0
RetailTrx|2026-11-03 00:00:00.0000000|1236
SdDocumentHeader|2026-11-03 20:36:01.1388966|0
ViewServerState|ViewDatabaseState|ShowPlan
---------------|-----------------|--------
0|0|0
P5
--
VIEW SERVER STATE denied: hardware + other-traffic rows go to the DBA checklist
Label|Ms|ResultRows
-----|--|----------
RetailTrx 1d by store|313|955
RetailTrx 7d by store|375|963
RetailTrx 92d by store|1078|972
RetailTrx 92d by city|954|203
RetailTrx 92d by channel x day|1157|1107
RetailTrxDetail 7d by store|907|965
RetailTrxDetail 92d by store|1813|974
SdDocumentHeader 1d by store x source|735|6777
SdDocumentHeader 7d by store x source|1532|12343
SdDocumentHeader 92d by store x source|1719|17591
SdDocumentHeader 92d by city x type|1517|1820
SdDocumentLog 7d by store x action|2908|5476
SdDocumentLog 92d by store x action|3220|9878
```

type	wayfinder-ticket
wayfinder	task
map	741
status	open
blocked-by	—

753 — The DBA adds the callout and donor-request tables to the replica publication

Question

v1's fourth subject area (callouts and donor requests, ticket 745) has no source: on 2026-11-03 the replica `172.23.27.47` / `POS` held no `Callout*`, `DonorRequest*` or `DeliveryCourierQuote` table. The owner is the DBA and will add them as transactional articles. This ticket is done when they are on the subscriber with recent rows.

HITL checklist (owner as DBA):

Must add. Every table has a primary key, as transactional replication requires. The schemas are BackOffice `Sartawi.Retail.Data/Modules/Callouts/Sql/` and `Modules/Pick/Sql/`. The exception is `DeliveryCourierQuote`, which is in `Modules/Sd/Sql/063_create_delivery_courier_quote.sql`.

Ready to run: 753-add-replica-articles.sql. It is a draft for the publisher with steps 0–4: - **0:** check `immediate_sync`, so the snapshot does not re-send `RetailTrxDetail`. - **1:** add the articles. - **2:** add the column filter. - **3:** refresh the subscription and snapshot. - **4:** run the count check.

Fill in the publication name.

- [ ] `Callout`: PK `CalloutId`. Includes `SearchRadiusKm` (script 007).
- [ ] `CalloutBid`: PK `BidId`.
- [ ] `CalloutEvent`: PK `EventId`.
- [ ] `DeliveryCourierQuote`: PK `DeliveryNo`. The dispatch scorecard joins it.
- [ ] `DonorRequest`: PK `RequestNo`.
- [ ] `DonorRequestLine`: PK `(RequestNo, DeliveryLine)`.
- [ ] `DonorRequestEvent`: PK `(RequestNo, Seq)`.

Optional dimensions, which are small: - [ ] `CalloutCourierStoreReason`: PK `Reason`. - [ ] `ServingList`, `ServingListStore` and `ServingListChange`.

Free text that may hold personal data. Your choice: column-filter these out of the articles. The semantic model never declares them either way. - `Callout.ReleaseNote` - `DonorRequestEvent.Reason` - `DonorRequestEvent.Detail` - `DonorRequestEvent.ComplaintRef` - `CalloutEvent.Detail`, added 2026-11-04. Script 005 made it NVARCHAR(300) because it "holds a pharmacist's or courier's words". - `ServingListChange.Note`, added 2026-11-04. It is NVARCHAR(500) human text and matters only if that optional dimension is added. - Consider `DeliveryCourierQuote.ErrorMessage` too: it is courier API error text.

Not tables: `DispatchBoardRelease`, which 750's missing list names, is only a screen authorization in BackOffice. Releases are columns on `Callout` (`ReleasedAt`, `ReleaseReason`).

Already on the replica, so no action: - `DeliveryHeader`, `DeliveryLine` - `SdDocumentLog`, `SdDocumentHeaderStatus`, `SdDocumentSource` - `Store`, `StoreArea`, `BusinessAddress`

Verify after the snapshot. Run this through the sanctioned `sqlcmd` shape:

```
SELECT 'Callout', COUNT(*), MAX(OpenedAt) FROM Callout
UNION ALL SELECT 'CalloutBid', COUNT(*), MAX(OfferedAt) FROM CalloutBid
UNION ALL SELECT 'DonorRequest', COUNT(*), MAX(RaisedAt) FROM DonorRequest;
```

Resolved when the articles are live and the counts match the publisher. Record: - the snapshot date; - whether the free-text columns were filtered; - the count check.

Progress

- **2026-11-04 00:18, subscriber check, not resolved.** The agent queried the replica through the sanctioned door; TCP 1433 was reachable. None of the 11 tables exist yet (`sys.tables` had no `Callout%` or `Donor%` match), so the articles have not been published.
- The session wrote the publisher script linked above. It found two free-text columns the first list missed and the correct `DeliveryCourierQuote` schema path. Both are folded into the body.
- The ticket goes back to `open`. Once the owner has run the script, any session can take it and run step 4's check on the subscriber.
- **2026-11-04 23:18–23:20, the articles are on the subscriber and replicating; ready to close.** The owner (acting DBA) published them from the server, and the snapshot landed 2026-11-04 23:16:08.
- **All 7 must-add tables exist,** with these row counts at 23:18: `Callout` 1,975, `CalloutBid` 12,724, `CalloutEvent` 83,505, `DeliveryCourierQuote` 2,861, `DonorRequest` 2,783, `DonorRequestEvent` 8,657, `DonorRequestLine` 3,283.
- **Live:** `CalloutEvent` went 83,505 → 83,509 → 83,517 over about 90 s. The latest `Callout` and `CalloutBid` rows are from 23:18:17, after the snapshot. The latest `DonorRequest.RaisedAt` is 23:06:45.
- **The optional dimensions were not added:** `CalloutCourierStoreReason` and `ServingList*`.
- **The free-text columns were not filtered.** `Callout.ReleaseNote`, `CalloutEvent.Detail`, `DonorRequestEvent.Reason`, `.Detail` and `.ComplaintRef`, and `DeliveryCourierQuote.ErrorMessage` are all on the subscriber. That is allowed: the semantic model never declares them, and 757's dbt models must not select them into any mart.
- **Owed to close:** the publisher-side counts for step 4. The sanctioned door reaches only the subscriber. Once the owner reports them, any session can close this.

Source: oms-react/.issues/assets/753-add-replica-articles.sql

```
/*
  753 — add the callout and donor-request tables to the POS replica publication.
  DRAFT for the owner (as DBA). Run on the PUBLISHER, in the published database. Read every step first.
  Map 741 · ticket .issues/753-the-dba-adds-the-callout-and-donor-request-tables-to-the-replica-
publication.md

  Step 0 asks one decision: whether the free-text columns are filtered out of the articles (the semantic
  model never declares them either way). The default below is FILTER, because no personal data should
  leave the publisher for an analytics subscriber.
*/

DECLARE @pub sysname = N'<<PUBLICATION NAME>>';   -- the publication DWPOSREPL03 / POS subscribes to

/* ------------------------------------------------------------------------------------------------
   Step 0 — check before touching anything.
   immediate_sync = 1 makes the next snapshot cover EVERY article (RetailTrxDetail is about 1.0B rows).
   If it is 1, turn it off first. allow_anonymous has to go off before immediate_sync can.
   That way the snapshot holds only the new articles.
------------------------------------------------------------------------------------------------ */
SELECT name, immediate_sync, allow_anonymous, status FROM syspublications WHERE name = @pub;
-- EXEC sp_changepublication @publication = @pub, @property = N'allow_anonymous', @value = 'false';
-- EXEC sp_changepublication @publication = @pub, @property = N'immediate_sync',  @value = 'false';

-- Every table must have a primary key (transactional replication refuses a table without one).
SELECT t.name, CASE WHEN EXISTS (SELECT 1 FROM sys.key_constraints k
                                 WHERE k.parent_object_id = t.object_id AND k.type = 'PK')
                    THEN 'PK' ELSE '** NO PK **' END AS pk
FROM sys.tables t
WHERE t.name IN (N'Callout', N'CalloutBid', N'CalloutEvent', N'DeliveryCourierQuote',
                 N'DonorRequest', N'DonorRequestLine', N'DonorRequestEvent',
                 N'CalloutCourierStoreReason', N'ServingList', N'ServingListStore', N'ServingListChange');

/* ------------------------------------------------------------------------------------------------
   Step 1 — add the articles. The seven required tables come first; the four optional dimensions are
   included and can be commented out.
------------------------------------------------------------------------------------------------ */
DECLARE @t TABLE (name sysname, required bit);
INSERT @t VALUES (N'Callout',1), (N'CalloutBid',1), (N'CalloutEvent',1), (N'DeliveryCourierQuote',1),
                 (N'DonorRequest',1), (N'DonorRequestLine',1), (N'DonorRequestEvent',1),
                 (N'CalloutCourierStoreReason',0), (N'ServingList',0), (N'ServingListStore',0),
                 (N'ServingListChange',0);

DECLARE @name sysname;
DECLARE c CURSOR LOCAL FAST_FORWARD FOR SELECT name FROM @t;
OPEN c; FETCH NEXT FROM c INTO @name;
WHILE @@FETCH_STATUS = 0
BEGIN
    IF NOT EXISTS (SELECT 1 FROM sysarticles a JOIN syspublications p ON p.pubid = a.pubid
                   WHERE p.name = @pub AND a.name = @name)
        EXEC sp_addarticle
             @publication       = @pub,
             @article           = @name,
             @source_owner      = N'dbo',
             @source_object     = @name,
             @destination_owner = N'dbo',
             @destination_table = @name,
             @type              = N'logbased',
             @pre_creation_cmd  = N'drop',
             @force_invalidate_snapshot = 1;
    FETCH NEXT FROM c INTO @name;
END
```

```
CLOSE c; DEALLOCATE c;

/* ------------------------------------------------------------------------------------------------
   Step 2 — vertical filter: leave the human-typed free text on the publisher.
   Includes two columns the ticket's first list missed (CalloutEvent.Detail and ServingListChange.Note,
   both NVARCHAR human text per BackOffice scripts Callouts/005 and 004).
------------------------------------------------------------------------------------------------ */
DECLARE @f TABLE (art sysname, col sysname);
INSERT @f VALUES (N'Callout',           N'ReleaseNote'),
                 (N'CalloutEvent',      N'Detail'),
                 (N'DonorRequestEvent', N'Reason'),
                 (N'DonorRequestEvent', N'Detail'),
                 (N'DonorRequestEvent', N'ComplaintRef'),
                 (N'ServingListChange', N'Note');           -- only if ServingListChange was added
                 -- (N'DeliveryCourierQuote', N'ErrorMessage') -- courier API error text; your call

DECLARE @art sysname, @col sysname;
DECLARE f CURSOR LOCAL FAST_FORWARD FOR SELECT art, col FROM @f;
OPEN f; FETCH NEXT FROM f INTO @art, @col;
WHILE @@FETCH_STATUS = 0
BEGIN
    IF EXISTS (SELECT 1 FROM sysarticles a JOIN syspublications p ON p.pubid = a.pubid
               WHERE p.name = @pub AND a.name = @art)
        EXEC sp_articlecolumn
             @publication = @pub, @article = @art, @column = @col, @operation = N'drop',
             @force_invalidate_snapshot = 1, @force_reinit_subscription = 0;
    FETCH NEXT FROM f INTO @art, @col;
END
CLOSE f; DEALLOCATE f;

/* ------------------------------------------------------------------------------------------------
   Step 3 — push the new articles to the existing subscription, then snapshot only those.
   sp_refreshsubscriptions covers subscriptions made with @article = 'all'. If DWPOSREPL03's
   subscription is per-article, use sp_addsubscription @article = <each name> instead.
------------------------------------------------------------------------------------------------ */
EXEC sp_refreshsubscriptions @publication = @pub;
EXEC sp_startpublication_snapshot @publication = @pub;
-- Watch Replication Monitor: the snapshot should list only the new articles.

/* ------------------------------------------------------------------------------------------------
   Step 4 — the count check. Run it on the PUBLISHER now, and the agent runs it on the subscriber
   (172.23.27.47 / POS) through the sanctioned read-only door. The counts should match within the
   replication lag.
------------------------------------------------------------------------------------------------ */
SELECT 'Callout' AS t, COUNT(*) AS n, MAX(OpenedAt) AS latest FROM Callout
UNION ALL SELECT 'CalloutBid',           COUNT(*), MAX(OfferedAt) FROM CalloutBid
UNION ALL SELECT 'CalloutEvent',         COUNT(*), NULL           FROM CalloutEvent
UNION ALL SELECT 'DeliveryCourierQuote', COUNT(*), MAX(CheckedAt) FROM DeliveryCourierQuote
UNION ALL SELECT 'DonorRequest',         COUNT(*), MAX(RaisedAt)  FROM DonorRequest
UNION ALL SELECT 'DonorRequestLine',     COUNT(*), NULL           FROM DonorRequestLine
UNION ALL SELECT 'DonorRequestEvent',    COUNT(*), NULL           FROM DonorRequestEvent;
```

type	spec
status	ready

3378 — HQ self-service analytics: the marts, the Cube model and the SIS.Api analytics doors (spec)

This is the **server half** of oms-react spec 766 (`C:\Playground\oms-react\.issues\766-hq-self-service-analytics-v1-spec.md`, status ready, settled 2026-11-05 for oms-react map 741). The stack is recorded in oms-react ADR 0006 (`C:\Playground\oms-react\docs\adr\0006-analytics-is-native-screens-over-cube-and-dbt-marts.md`).

766 is the authority on **what** is built. Its sections A–F are cited here by letter and are not re-argued. **This spec becomes the normative owner of every server shape** that 766 §C proposes. If a ticket here changes a shape, record it in that ticket's Comments, and oms-react corrects its transcription. 766's drift rule applies.

The web half is oms-react 767–784, built against stubs of this wire. oms-react 784 is the live walk on staging, and it waits on 3381–3383.

Problem Statement

People at HQ get answers about till sales, orders by channel and delivery speed only by asking a developer, who runs SQL against the POS replica `DWPOSREPL03`. The answers drift from the copilot's and the dispatch scorecard's figures. Nothing can be saved, shared or refreshed on its own.

The replica must not carry interactive analytical load, and nothing may be written into `POS`.

Solution

- **dbt** builds incremental daily marts and one-row-per-item fact tables into `POS_Analytics`. That database already exists, beside `POS`, with its logins.
- **Cube Core** declares the semantic model once over those marts and is the only query engine.
- **SIS.Api** adds:
- a gate (`Analytics/Access`);
- the model door (`Analytics/Model`, `Analytics/DimensionValues`);
- one governed query door (`Analytics/Query`), which validates, compiles, caps, caches and audits;
- a store for analytics views and boards, with tiers and revisions;
- three tools in the copilot catalogue.
- **All of it runs as one container image on infra's existing Nutanix container platform** (oms-react 749, amended 2026-11-05; it was "one Docker host VM"). SIS.Api is the only caller.

> **Amendment 2026-11-05 (owner, grilled in oms-react).** This overrides the hosting lines below (images, secrets, the overlap guard, the compose file): - **One image** (`oms-analytics:<version>`): Cube API with `CUBEJS_REFRESH_WORKER=true`, Cube Store bound to localhost only, and the dbt jobs with their 15-minute / 03:00 schedule inside, under a small process manager. All logs go to stdout. - **Exactly one instance**, no persistent disk (Cube Store is a cache and rebuilds on restart). - **The overlap guard is a lock file** (`flock`) inside the container, not the fixed container name. There is no Docker socket on infra's platform, so the scheduler-launches-a-container design goes. - **Secrets:** three env vars set by infra at deploy

time (`CUBEJS_DB_PASS`, `DBT_ENV_SECRET_PASSWORD`, `CUBEJS_API_SECRET`), never in the image. - **TLS in:** infra's platform terminates 443 with a Cloudflare certificate on an internal-only name → container port 4000. **TLS to the replica:** pinned to its current certificate (D8 (a)), never `trustServerCertificate`. - **Health:** infra's platform probes Cube's `/readyz` and `/livez`. - The infra ask is oms-react `docs/research/741-nutanix-infra-request.md` (rewritten 2026-11-05).

User Stories

1. As the analytics screens, I want one access probe that says whether the user may view and whether they may curate, so that I can hide what they cannot use.
2. As the analytics screens and the AI, I want one catalogue of measures and dimensions with their shape, unit and labels in English and Arabic, so that neither of us hard-codes the model.
3. As the analytics screens, I want to search store, city and region values, so that the pickers stay fast.
4. As the analytics screens, I want one query door that takes a saved view or an inline definition plus board filters, and returns rows, the resolved period and freshness, so that every panel is one call.
5. As the business, I want every query validated against the model and the chart-shape rules before it runs, so that no invalid or PII query reaches Cube.
6. As the business, I want each relative period resolved on the server in Riyadh time, with Sunday weeks, capped at today, so that "last 30 days" always means the same thing.
7. As the business, I want every query capped at 30 s and 10,000 grouped rows, with `truncated` reported, so that no question loads the server.
8. As the business, I want a percentile measure refused over more than 92 days, so that `fct_delivery` scans stay bounded.
9. As the business, I want ten screens on one board to cost about one Cube query per interval, so that wall screens are cheap.
10. As the business, I want Cube's errors mapped to `ANALYTICS-*` codes and scrubbed, so that no engine internals reach a client.
11. As an HQ user, I want my views and boards stored with their owner, tier and revisions, so that my work is kept and can be shared.
12. As an HQ user, I want editing an item I may not edit refused with a code that says "save as copy", so that I never overwrite someone else's work.
13. As an HQ user, I want a save based on a stale revision refused, naming who saved last, so that neither of us loses work.
14. As a curator, I want to certify a revision into a frozen copy, and to un-certify or delete it, so that the trusted figures are under control.
15. As an editor, I want the last 20 revisions of an item and a revert that is itself a revision, so that a bad save is never final.
16. As the AI, I want `describe_model`, `semantic_query` and `create_view` in the one catalogue, so that I answer from the same figures as the screens.
17. As the business, I want MCP to see only the two read tools, so that MCP stays read-only.
18. As the business, I want every query and every store write audited with the acting user, so that we know who asked what.
19. As a DBA, I want the replica to see only dbt's bounded, scheduled `SELECT`s, and a stale source to fail a run closed, so that analytics never competes with replication and never shows a gap.
20. As the business, I want `sales_summary`, `orders_by_channel` and `callout_stats` to read the semantic model at parity, and the interim `TillSalesStoreDay` rollup retired, so that there is one number for each thing.

Implementation Decisions

- **Everything in 766 §A–§D holds,** including the provisional values and their 750 triggers.
- **The `Analytics/` tree** sits at the repo root, beside the solution, with `dbt/` and `cube/`.
- **dbt:** dbt-core with `dbt-sqlserver[pyodbc]`, pinned. 757 proved 1.12.6 and 1.12.2.
- **Cube:** Cube Core, pinned. 756 proved v1.8.18.
- **Images:** pinned images, delivered from the internal registry or as offline tarballs (oms-react's `741-nutanix-infra-request.md` §4).
- **Secrets:** the host's env files only, never the repo.
- **The marts** follow 766 §B and oms-react 748:
- the daily marts per area, and `fct_delivery`;
- shares stored as a numerator and a denominator;
- `_loaded_at` on every row, and `mart_load_log`;
- the `intraday` job (every 15 minutes, swap in one transaction) and the `nightly` job (03:00, 3-day rebuild, `dbt test`, reconciliation);
- `dbt source freshness` run first, as the fail-closed gate, with the UTC override;
- the fixed container name as the overlap guard.
- **The Cube model** follows 766 §B and oms-react 756:
- every cube private and only views public;
- `meta` on each member with `shape`, `unit`, `area`, `maxWindowDays?` and `label` and `description` in English and Arabic;
- the custom Sunday week;
- `CUBEJS_DB_QUERY_LIMIT` and `CUBEJS_DB_QUERY_DEFAULT_LIMIT` both 10000, `CUBEJS_DB_QUERY_TIMEOUT` 30 s, and `total` always sent;
- TLS through a `driverFactory` pinned to the replica certificate, never `trustServerCertificate`;
- a `refreshKey` on the newest good run in `mart_load_log`, checked every 30 s;
- no pre-aggregations in v1.
- **The SIS.Api doors** are 766 §C's wire. This spec owns it from now on:
- `Analytics/Access`, `Model`, `DimensionValues` and `Query`;
- the `Analytics/Item` family;
- the codes `ANALYTICS-401` to `410`.

SIS.Api reaches Cube at `https://<analytics-host>/cubejs-api/v1` with a service credential (a signed Cube API token). Before `/load`, it checks every member against `/meta`. - **The validator and compiler** form one module. It takes a stored or inline definition and the board filters, and returns a resolved period plus a Cube `/load` query, or an `ANALYTICS-406/407` refusal. The chart-shape table is 755 Q3 with 766 §E's two amendments: the combo has one axis per pane, and a breakdown has at most 5 series plus "Other". The **same module** serves the query door and `create_view`. - **The caches:** - Cube's own cache, keyed on the `refreshKey`; - SIS.Api's 30 s cache, keyed on the canonical compiled query and the resolved period; - single-flight coalescing of identical in-flight queries.

The concurrency cap toward Cube is 8 (config). - **The store** lives in **SIS.Api's own database on HQ**, never in `POS_Analytics`. It holds an item table and a revision table with the full definition snapshot of each revision, keeping the last 20 per item. Certification writes a frozen copy that records `certifiedFrom`. The tiers and the curator grant are 755 Q1. - **The grants:** a new `BackOfficeScreen` grant for Analytics (view), and a curator grant (working name `AnalyticsCurate`). This spec's first door ticket names both. - **The copilot tools** are 766 §D: - `describe_model`, `semantic_query` and `create_view` join the one registry; - the registry gains a surface tag, with `create_view` on `panel` only; - the D10 answer gains `viewRefs[]`; - the D14 set gains Q1–Q15. - **The audit:** every query and every store write records the user, the item, the shape of the compiled query, the rows and the milliseconds. AI calls also write 3348's audit record.

Testing Decisions

- **Test only external behaviour:** the rows and codes a client gets, the `/meta` contents, the store's state, and the objects in `POS_Analytics`. Never internal calls.
- **dbt** (in `Analytics/`):
- `dbt test` with uniqueness and not-null;
- the till-value reconciliation against a direct `RetailTrx` sum for yesterday;
- the source-freshness gate exiting non-zero with no swap;
- for area 4, scorecard parity with `DispatchScorecardMath`.
- **Cube:** a contract script over `/meta` checks that no PII member exists, every cube is private, and every public member has a shape, a unit and both labels. It runs before every deploy.
- **SIS.Api** (`Data.Tests/Analytics`):
- golden tests of definition → `/load` for the 15 evaluation questions;
- validator tests over the shape table;
- period resolution: Riyadh time, Sunday weeks, capped at today;
- code mapping for Cube's error statuses;
- single-flight;
- the store's permissions per tier, the revision conflict, certify-freezes-a-copy, revert, and the 20-revision trim;
- the audit, per door.

Prior art: `Data.Tests/Copilot` (3346, 3348). - **The AI:** the extended D14 set runs on staging before any prompt, model, tool-description or semantic-model change ships.

Out of Scope

Everything in 766's Out of Scope. In particular: - Metabase in any role; - SQL or arbitrary joins; - row-level security; - writes to `POS`, and SQL Agent jobs on `DWPOSREPL03`; - an admin screen for the audit.

Further Notes

- **Tickets:**
- 3379: the marts;
- 3380: the Cube model;
- 3381: the query doors;
- 3382: the store;
- 3383: the copilot tools;
- 3384: area 4;
- 3385: parity, and retiring the interim rollup.
- **Gates:**
- 3384 waits on oms-react 753, the DBA adding the callout and donor-request articles to the replica publication.
- oms-react 750 (the replica headroom probe) gates no ticket. It only confirms the provisional values.
- **Build inputs:** the host name, its IP and SIS.Api's source IPs come from infra (oms-react 749). 3379 records them.

Source: `BackOffice/.issues/3379-the-analytics-host-builds-the-till-order-and-delivery-marts-into-pos-analytics-every-15-minutes.md`

status	open
spec	3378
blocked-by	—

3379 — The analytics host builds the till, order and delivery marts into POS_Analytics every 15 minutes

What to build

The first analytics data, end to end on the real host. The `Analytics/` tree is created with `dbt/` and `cube/` (spec 3378). The dbt project builds areas 1–3 of the semantic model (oms-react 766 §B and `741-v1-semantic-model.md`) into `POS_Analytics`:

- **Till sales:** `mart_till_sales_daily`, by date × store × country × NPHIES-linked flag. It holds additive counts and SAR sums, and shares as numerator and denominator.
- **Orders:** `mart_orders_daily`, by date × store × channel group × document type.
- **Delivery speed:** `fct_delivery`, one row per delivery with its first-ever `DCRT`/`DRDY`/`DOFD`/`DDLR` times (ADR 0081), plus `mart_delivery_daily`.
- **The dimensions** are dbt seeds and models. The `DocumentSource` → five-group map is a seed and the only copy of it. Weeks start on Sunday, and dates are capped at today.
- **`_loaded_at` on every row,** and `mart_load_log` with one row per run, recording each source's newest event time.
- **The jobs.** `intraday` runs every 15 minutes: today's slice, plus yesterday's before 02:00, swapped in one transaction. `nightly` runs at 03:00: the last 3 days, then `dbt test` and the till-value reconciliation. Each runs `dbt source freshness` first (with the UTC override), and a stale source fails it closed with a logged row. The fixed container name is the overlap guard.
- **The host's compose file:** the Cube API, the Cube refresh worker, Cube Store and the dbt scheduler. The images are pinned, and the secrets live in the host's env files only.

Spine reach

data (dbt marts, seeds, tests) / ops (compose, scheduler, env)

Proof (→ `tdd` red-green cycles)

- [x] `dbt test` passes: unique and not-null keys on every mart, and the channel seed covers every `DocumentSource` seen in the last 92 days · dbt *(proved locally on synthetic data; on real sources it is owed, see Comments)*
- [x] `till_value_reconciles` — yesterday's `mart_till_sales_daily` net SAR equals a direct `RetailTrx` sum to the halala · dbt singular test
- [x] `stale_source_fails_closed` — with the source's newest event forced old, the job exits non-zero, no swap happens, the last slice is intact and `mart_load_log` records the failure · job script test
- [x] `intraday_swap_is_atomic` — a reader under RCSI during a killed swap sees the old day, never a gap (757's method) · local spike repeated on the host *(proved locally only — the host run is owed, see Comments)*

Boundaries

- Read-only on `POS`, through `svc_analytics_dbt`: bounded, date-filtered `SELECT`s only, one or two threads.
- Nothing is written outside `POS_Analytics`.
- No SQL Agent jobs.
- This ticket records the **build inputs**: infra's host name and IP, and SIS.Api's source IPs.

Done when

Both jobs run on the analytics host on schedule, all four tests are green, and `mart_load_log` shows good intraday runs for a full day.

Blocked by

None. It needs the VM from infra (oms-react 749); until then it runs against a local SQL Server 2022, as 757 did.

Comments

2026-11-05 — built and proved locally; the host half is owed (status stays open)

Built (branch `t3379`), in a new `Analytics/` tree at the repo root: - `dbt/`: the dbt project (dbt-core 1.12.6, `dbt-sqlserver[pyodbc]` 1.12.2, every transitive dependency pinned, 757's set), its job image (`python:3.12-slim-bookworm` pinned by index digest, `msodbcsql18` 18.7.1.1-1, `TZ=Asia/Riyadh`, the replica certificate copied in from `dbt/certs/`) and `run-job.sh`, the job entrypoint. - **Marts:** `mart_till_sales_daily`, `mart_orders_daily`, `fct_delivery` and `mart_delivery_daily`. - **Dimensions:** `dim_date` (Sunday weeks, rows only up to today) and `dim_store` (StoreArea City and RegionAreaName, the scorecard's join). - **Seeds:** `channel_map` (the only copy of the five-group map), `channel_groups` and `loyalty_placeholder_ids`. - **The log:** `mart_load_log`, plus the `mart_freshness` view, which holds "data as of" and Cube's refreshKey. - Every mart is incremental `delete+insert` and carries `_loaded_at`, `_source_newest_at` and `_invocation_id`. - `scheduler/`: busybox crond plus the Docker CLI, at `TZ=Asia/Riyadh`. - `*/15` runs intraday and `0 3` runs nightly. - `launch.sh` starts each job as the fixed-name container `analytics-dbt`, which is the overlap guard. - When the name is taken, an intraday run skips. The nightly run waits for up to 30 minutes, so it is never lost to an intraday run in flight. - `compose.yaml`: Cube API, the Cube refresh worker, Cube Store and the dbt scheduler, plus the dbt job image under profile `job`. - The images are pinned (Cube v1.8.18 and Cube Store by 756's digests). - The secrets are only in `env/*.env`, which is excluded from the repo. `env/*.env.example` carries the keys. - `cube/`: an empty placeholder for 3380. - `local/`: the local-only stack (SQL Server 2022 CU27 in Docker, `TZ=Asia/Riyadh`, compat 150, RCSI on, the two `svc_analytics_*` logins with 749's grants, and a synthetic POS), plus the three proof scripts.

Proved locally (synthetic POS: 60 stores, 180k till rows, 36k orders, 28.8k deliveries, 116k log events; never the replica; dbt ran as `svc_analytics_dbt`, which holds only `db_datareader` on POS): - **`dbt test`: green.** A nightly `dbt build` passed 49 of 49 nodes. - Every test was mutation-checked by `local/prove-tests-bite.sh`, and each one **fails** on data that breaks it: - a halala off on yesterday fails `till_value_reconciles`; - an order from an unmapped code `ZZZZ` fails `channel_map_covers_recent_sources`; - a duplicated key row fails `unique_key`; - a NULL store fails `not_null`. - That check caught a real gap: the marts' tests had never been written to `marts.yml`. - **`stale_source_fails_closed`: PASS** (`local/test-stale-source.sh`). With SdDocumentLog's last 90 minutes moved aside and 200 new sales landed today: - the job printed `ERROR STALE` and exited **3**; - today's slice stayed at 1682, and no success row was written; - one `stale_source` row named `SdDocumentLog`; - after the events were restored, the next run landed the 200 sales (1882). - **`intraday_swap_is_atomic`: PASS** (`local/test-swap-atomic.sh`). - **How the swap was caught open:** sa held `mart_load_log` locked, so the real job blocked on its own load-log post-hook. That point is inside the swap transaction, after the delete+insert and before COMMIT, so

the test needs no test code in the project. A NOLOCK read at that moment showed the uncommitted new day (2522). - The job's container was then `docker kill`ed. - The `svc_analytics_cube` reader took **162,357 samples and every one saw the old day (2222)**. The longest read was 42 ms. - After the kill the old day stood, with no success row and no open session. The next run landed 2522. - **Logic checks:** - A delivery that re-posted DOFD and DDLR a day later kept its first times (ADR 0081). - Each delivery-minute figure matched hand arithmetic: 144 and 105. - 2028-dated till rows reached no mart. - The 2026-11-05 row has `week_start` 2026-11-01, a Sunday. - Before 02:00, intraday rewrote yesterday as well (`first_day_written` 2026-11-04). - **The scheduler, locally** (the scheduler container on the Docker socket): - cron fired intraday at 01:45 Riyadh by itself; - with `analytics-dbt` already taken, an intraday launch logged "skipped (overlap guard)"; - a nightly launch waited for the name to free, then ran green. - `compose.yaml` validates (`docker compose config`). - The job image also builds for **linux/amd64**, the host's arch (757 had built only arm64), and an amd64 intraday run went green under emulation. - **Write audit:** after every run, POS held only the seeded tables. The dbt login has no permission on POS beyond `db_datareader`.

Decisions made here (shape changes for oms-react to transcribe, 766's drift rule): - **Loyalty placeholders** are the union of 741's list (3795407, 3378, 1000000004) and BackOffice's `LoySentinelMembers` (3795407, 2812732). They are kept in a seed. - **Till value is `Amount x rate` (741).** It is not `AmountValue`, which 3346's `TillSalesStoreDay` reads, so 3385's parity check must expect that difference or settle it. - **Mart columns:** - the till mart also stores `returns_value_sar`, `net_till_value_sar` and `nphies_linked_sales`, all additive; - money is `decimal(19,3)` (sql-create's 3-decimal rule); - the marts are clustered columnstore and the dimensions are rowstore. - **Delivery definitions:** - "created" is `DeliveryHeader.EntryTime`, as 2639 rules; - the order is `RefDocumentNo`; - the channel is the order's DocumentSource, else the delivery's `RefDocumentSource`; - a negative span has no time. - **`mart_delivery_daily`** rebuilds the slice's days plus the created date of every delivery that `fct_delivery` re-derived in the same invocation. A delivery created last week and delivered today therefore moves last week's count. - **Intraday runs `dbt run` (models only).** The seeds and every test belong to the nightly's `dbt build`, so the 92-day coverage read never runs every 15 minutes. - **Freshness:** - the UTC override is documented at `macros/freshness.sql` as the ticket's named exception (elapsed-time math only); - each source's freshness `filter` also bounds the read to 2 days and drops future rows; - every business date reads the server's local clock (`SYSDATETIME()`), with no `AT TIME ZONE`.

Owed: 1. **The run on the real host.** That covers four things: - loading the images offline; - the replica certificate in `dbt/certs/`; - `env/*.env` at mode 0600; - `compose up`, then the four proofs again against `DWPOSREPL03`, with the swap-kill on the host. 2. **A full day of good intraday runs** in `mart_load_log` on the host. Watch for false `stale_source` at night: the 60-minute `error_after` is 757's value, and the quiet hours on real traffic are unmeasured. 3. **The channel seed's insurance-platform code(s).** 717 counted 17.5% of orders under the group but did not record the code. The first nightly on the host lists every unmapped code, and those rows are then added to `channel_map.csv`. 4. **The build inputs: 749 did not supply them.** They are infra's host name and IP, and SIS.Api's source IPs. They go into `env/*.env` and this ticket when infra hands the VM over. 5. **Not this ticket's (noted for 3380 and 3381):** the 443 TLS front that SIS.Api calls, and Cube's TLS `driverFactory`.

3378 review (2026-11-05, branch `t3378-review`): the freshness gate. - **Decision: the gate's thresholds are per job.** Intraday keeps 30/60 minutes per source (757). The 03:00 nightly uses **180/360**: every source must have delivered something since 21:00 the evening before. With the old single 60-minute gate, an overnight quiet hour (no delivery or order after 02:00) skipped the nightly, its tests, the reconciliation and the 3-day rebuild, night after night, with nothing wrong. A replica that stopped before the evening still fails the nightly closed. The vars are `freshness_warn_minutes`/`freshness_error_minutes` (defaults in `dbt_project.yml`, each job's own passed by `run-job.sh`). Considered and not taken: a replication-heartbeat gate (`MSreplication_subscriptions` on the subscriber). It is activity-independent, but whether the Distribution Agent stamps it while idle is unverified, and it cannot be proved locally. Revisit it if 360 minutes misfires on real traffic. That is owed item 2 above, which now covers both windows. - **No UTC in the gate (no-utc-time).** dbt-adapters 1.24.5 computes the age as `snapshotted_at - max_loaded_at` and only tags naive values UTC (`_utc`: `replace(tzinfo=UTC)`, no shift). So

the snapshot is now `SYSDATETIME()` (local), every `loaded_at_field` is its raw column, and the filters use `SYSDATETIME()`. The "UTC override" this ticket named is gone, together with every `SYSUTCDATETIME()` and 3-hour `DATEADD`. On the host, check once that `SELECT SYSDATETIME()` on the replica is Riyadh time (the business dates in `time.sql` assume it already). - **Proved on the local stack:** `stale_source_fails_closed` PASS (fresh sources PASS at age ~0, so the naive-local age is right) and the new `local/test-nightly-quiet-hour.sh` PASS. With every source silent for 90 minutes, intraday exits 3 and the nightly runs green (49/49). With SdDocumentLog silent for 7 hours, the nightly exits 3 with a `stale_source` row. Both scripts now count only their own run's log rows (`T0`), and the stale test's 200 sales carry a time-unique key, so either script can run again on the same day.

2026-11-05 — hosting amended: one image, no VM (owner, grilled in oms-react 749)

Infra runs our apps as containers on Nutanix, so there is no VM, no compose file and no Docker socket on the target. Spec 3378 carries the amendment. **What this ticket now owes, in addition:** 1. **Repackage into one image** `oms-analytics:<version>` (linux/amd64): Cube API with the refresh worker on, Cube Store on localhost, the dbt project, and an in-container scheduler (supercronic or busybox crond, `TZ=Asia/Riyadh`) under a process manager. `compose.yaml` stays as the **local** harness only. 2. **The overlap guard becomes** `flock -n /tmp/analytics-dbt.lock run-job.sh …`. A skipped launch still logs "skipped (overlap guard)". Re-prove the overlap test inside the one container. 3. **Logs to stdout,** each job line prefixed `dbt:` and carrying the run id and its exit code, so infra's log viewer is enough to see a run. 4. **Secrets from env vars** only; `env/*.env` remains for the local harness. 5. Owed item 1 reads "deploy the image through infra" instead of "`compose up` on the host", and owed item 4's build inputs are now infra's **internal DNS name** and SIS.Api's source IPs (no host IP).

2026-11-05 — the one image built and proved locally (branch `t3379-one-image`, worktree `BackOffice-analytics-image`)

Items 1–4 above are done in `Analytics/image/` (README there): `Dockerfile` on the pinned Cube v1.8.18 base (Debian 13) with Cube Store's binary, dbt in a venv (Python 3.13), msodbcsql18 18.7.1.1, supercronic 0.2.51 (sha-pinned); `supervise.sh`, `job.sh` (`flock` guard), `front.js` (`/health`), `deploy.py`, `crontab`, `deploy/` (k8s + compose for infra). Non-root (uid 10001). Added: `run-job.sh rebuild` and the **deploy check** — a changed dbt project (hash in `dbo.analytics_deploy`) full-rebuilds the marts once on start.

`local/test-one-image.sh` — **10/10 PASS** (amd64, emulated): missing secret → exit 64 · ready through 4000 · uid 10001 · crontab valid · Cube Store loopback only · first start full rebuild + recorded (pinned TLS, no trust-cert) · `/health` 200 fresh · `/load` 200 with a token, 403 without · intraday skipped while nightly runs · restart does not rebuild · a dying process exits the container · `/health` 503 stale past the threshold. Found and fixed on the way: `autoremove` dropped `libgssapi-krb5-2` (msodbcsql18 needs it, undeclared) — the build now fails on any missing native library. Still owed: the run on infra's platform against `DWPOSREPL03` (item 1), and a day of good runs (item 2).

status	open
spec	3378
blocked-by	3379

3380 — Cube answers only declared members over the marts, with bilingual labels and shape metadata

What to build

The Cube model for areas 1–3, over the 3379 marts (oms-react 766 §B, 756):

- **Every cube is `public: false`,** and only views are public, so `/meta` is exactly the catalogue.
- **Every public member's `meta`** carries `shape` (additive, rate or percentile), `unit`, `area`, `maxWindowDays` (92 for percentiles) and its `label` and `description` in **English and Arabic**.
- **Rates are Σ numerator ÷ Σ denominator** over the filtered set. Percentiles are computed exactly per query with `PERCENTILE_CONT` over `fct_delivery`.
- **A custom Sunday week granularity.**
- **The limits:** `CUBEJS_DB_QUERY_LIMIT` and `CUBEJS_DB_QUERY_DEFAULT_LIMIT` are both 10000, and `CUBEJS_DB_QUERY_TIMEOUT` is 30 s.
- **TLS** through a `driverFactory` pinned to the replica's certificate, never `trustServerCertificate`.
- **The cache:** a `refreshKey` on the newest good run in `mart_load_log`, checked every 30 s. No pre-aggregations.
- **Area 4 is declared unavailable,** and its measures are not exposed until 3384.

Spine reach

model (Cube schema, config) / ops

Proof (→ `tdd` red-green cycles)

- [x] `meta_contract` — over `/meta`: no PII member (no customer, member, staff, product or free-text dimension), every cube private, and every public member has a shape, a unit and both labels · contract script, run before each deploy
- [x] `fifteen_questions_answer` — Q1–Q10, Q14 and Q15 each answer through `/load` with the expected members · script against the host
- [x] `percentile_matches_scorecard_math` — p50 and p90 by store over 92 days equal `DispatchScorecardMath.Percentile` on the same rows · parity script

Boundaries

Cube's login is `svc_analytics_cube`, a `db_datareader` on `POS_Analytics` only. The Cube API is not reachable from browsers (F4 of the infra request).

Done when

The contract script and both parity scripts pass against the host, and `/meta` lists exactly the area 1–3 members.

Blocked by

3379

Comments

2026-11-05: built, and all three proofs pass locally; the host run is owed (status stays open)

The proofs are ticked because they ran and passed against the **local** stack. Done-when also needs them to pass against the host, so that part is still owed.

Built (branch `t3380`), in `Analytics/cube/` and the Cube services of `compose.yaml`: - **`model/cubes/`**: every cube is private. - `till_sales_daily`, `orders_daily` and `fct_delivery` (JS, so the percentile SQL is generated once); - `stores` (dim_store), `calendar` (dim_date) and `mart_freshness`. - Each `refresh_key` reads `mart_freshness` (newest good load id | data as of) every 30 s. There are no pre-aggregations. - **`model/views/`**: the only public members. - `till`, `channel_orders` and `delivery` (areas 1–3); - `callouts` (area 4: `available: false`, a bilingual `unavailableReason`, no measure); - `freshness` (a system view, see the decisions below). - The shared dimension keys are aliased the same way in every view: `date`, `store`, `store_name`, `city`, `region`, `country` (till only) and `channel` (orders and delivery only). - **Meta on every published member**: - measures carry `shape`, `unit`, `area`, the label and description in en and ar, and on a percentile `maxWindowDays: 92` and `percentile`; - dimensions carry `kind` and the labels, and `channel` and `country` carry their `values` inline with labels; - views carry `area`, `available`, `label` and `description`. - Rates are Σ numerator ÷ Σ denominator; the numerators are private sums. `sunday_week` (origin 2024-01-07, a Sunday) is on every time dimension. - **`cube.js`**: the TLS `driverFactory`. - It sets `encrypt` and `trustServerCertificate: false`, and pins the server certificate to `cryptoCredentialsDetails.ca` = `certs/db-ca.pem` (`ANALYTICS_DB_CA_FILE`), with the host name checked (`ANALYTICS_DB_TLS_SERVERNAME` when needed). - A missing file throws, so the connection fails closed. - **`compose.yaml`**: `TZ=Asia/Riyadh` and `ANALYTICS_DB_CA_FILE` added. The limits (10000/10000) and the timeout (30s) were already 3379's and are proved live here. - **`proofs/`**: - `meta_contract.mjs`: run before every deploy. It checks the exact member list, every cube private, no PII, product or free-text dimension, shape, unit, area and labels, the area-4 declaration, and `sunday_week`. - `fifteen_questions_answer.mjs`. - `percentile_matches_scorecard_math.ps1`. - **`local/cube.sh`** and `cube.Dockerfile`: the local stack. It runs on 3379's SQL container, which now offers TLS with a self-signed certificate (offered, not forced).

Proved locally (3379's synthetic marts, never the replica; Cube ran as `svc_analytics_cube`): - **`meta_contract`: PASS.** It was mutation-checked: a public `stores` cube, a missing Arabic label and a `customer_mobile` dimension each fail it (5 FAILs). - **`fifteen_questions_answer`: PASS.** - Q1–Q10, Q14 and Q15 answer through `/load`, 200, with every expected member in every row and `total` present. Each took 12–550 ms. - Q11–Q13 are refused (400 `not found`). - The query sends explicit local dates, no `timezone` and `total: true`. Weeks start on Sunday. - `return_rate` = Σreturns ÷ Σsales (0.055811), which is not the mean of the store rates (0.056133). - A limit of 10001 is refused, and `freshness` answers one row. - **`percentile_matches_scorecard_math`: PASS.** - Over 92 days (26,517 deliveries), 240 figures by store (60 stores × p50/p90 × both spans) equal the real `DispatchScorecardMath.Percentile`, loaded from the built Sartawi.Retail.Data, and T-SQL `PERCENTILE_CONT OVER (PARTITION BY)`. The worst gap was 1.7e-13 min. - By store × day over 7 days, 1,672 figures equal it, 404 of them interpolated between two different values. - The estate totals match too. - The delivery counts and the same-day share equal `mart_delivery_daily` for all 60 stores. - It was mutation-checked: `PERCENTILE_DISC` fails it. - **TLS:** - pinned to the right certificate, `sys.dm_exec_connections` shows `encrypt_option = TRUE` for `svc_analytics_cube`; - pinned to a different self-signed certificate, the connection is refused (`self-signed certificate`), and Cube stays up, answering 400; - a wrong host name is refused (`Hostname/IP does not match certificate's altnames`); - a missing CA file is refused (`ENOENT`). - **Auth:** no token and a bad signature each get 403.

Decisions and shape changes (for oms-react to transcribe under 766's drift rule): - **Percentile SQL.** T-SQL `PERCENTILE_CONT` is a window function only, so it cannot sit under Cube's GROUP BY. Each percentile measure is `(SELECT TOP 1 PERCENTILE_CONT(p) … OVER () FROM OPENJSON('[' + STRING_AGG(v, ',') + ']'))`, an outer-reference aggregate, which still *is* `PERCENTILE_CONT`. - A first design used ordered `STRING_AGG … WITHIN GROUP`. The parity proof found that design breaks any query that asks both delivery spans ("mutually incompatible orderings"), so it was replaced. - **This needs SQL Server 2017+ and `POS_Analytics` at compatibility level 130 or higher.** - **Area 3 is one cube over `fct_delivery`,** counts included, so every area-3 measure can share one query. The counts equal `mart_delivery_daily`'s, which the proof checks. The daily mart is unused by Cube for now. - **Percentile keys:** `order_to_delivered_min_p50` / `_p90` and `ready_to_delivered_min_p50` / `_p90`. 741 named them as one measure "(median, p90)". - **Area 4 is a `callouts` view with one member,** the calendar `date`, because Cube drops a view that has no members from `/meta` (proved). So `/meta` lists the area 1–3 members plus that declaration and the system view. - **`dataAsOf`** comes from the **system view `freshness`** (`meta.system: true`, no area). Cube's `/load` does not return refreshKey values (checked), and SIS.Api has no login on `POS_Analytics`. SIS.Api leaves it out of `Analytics/Model`. - **Breakdowns** are the measure's own view's dimensions. There is no per-measure list in meta. - **The 92-day cap is meta only** (`maxWindowDays`). SIS.Api enforces it with `ANALYTICS-407`, and Cube does not.

Owed: 1. **The host run.** - Put the replica certificate at `cube/certs/db-ca.pem`; its name must match `CUBEJS_DB_HOST`, or set `ANALYTICS_DB_TLS_SERVERNAME`. - Fill `env/cube.env`, then run `compose up`. - Run the three proofs against `DWPOSREPL03`'s `POS_Analytics` (`MIN_ROWS` for quiet filters such as Riyadh HungerStation). - Only then do Done-when and `status: done` hold. 2. **The host's SQL Server version and `POS_Analytics` compatibility level** (≥ 2017 / ≥ 130) for the percentile SQL, and a timing of a 92-day estate percentile on real volumes (about 1.4M deliveries) for 750. 3. **`compose config` was not re-run here,** because this shell has no compose plugin. The YAML parses, and the five services are intact. 4. **Arabic wording review by the owner** (766 §B: the owner approves any new or reworded measure). This includes "طلبات الاستدعاء" for callouts. 5. **Not this ticket's:** the 443 TLS front (749), and 3384's area-4 cubes.

Amended 2026-11-05 (spec 3378 amendment): Cube ships inside 3379's single image, not as two compose services. Run the API with `CUBEJS_REFRESH_WORKER=true`, and point it at Cube Store on localhost. The 443 front is infra's platform (Cloudflare certificate, internal-only name → port 4000). The replica TLS stays pinned (D8 (a)). Owed item 1's "`compose up`" becomes "deploy the image through infra", and item 3 (compose config) applies to the local harness only.